

TBU

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March 30, 2026

1 Set Up

```
> # Install remotes if you don't have it
> if (!require("remotes")) install.packages("remotes")
>
> # Install the micd package from the BIPS repository
> remotes::install_github("bips-hb/micd")
>
> # Load the library
> library(micd)
> library(pcalg)
```

```
> library(tidyverse) #installed tidyverse
> library(nlme)
> library(ggplot2)
> library(here)
> library(kableExtra)
> library(modelsummary)
> library(dplyr)
> library(performance)
> library(glmnet)
> library(pcalg)
> library(gridExtra)
> library(bnlearn)
> library(igraph)
> library(gt)
> library(gtExtras)
> library(corrplot)
> library(ggpubr)
> library(dagitty)
>
> bootstrap_sample_size <- 1000
>
> set.seed(516)
> getwd()
```

```
[1] "/Users/sunilmd/Desktop/vscode/MATH-516/project-3-nikita-zhuo/ANALYSIS_TESTING_1ALPHA_2OPTA"
```

```
> data <- read.csv(here("low2076upd.csv"))
> colSums(is.na(data)) #no NAs in any columns
```

Y	A	LIMIT_BAL	SEX	EDUCATION	AGE	PAY_0	PAY_2
0	0	0	0	0	0	0	0

PAY_3	PAY_4	PAY_5	PAY_6	BILL_AMT1	BILL_AMT2	BILL_AMT3	BILL_AMT4
0	0	0	0	0	0	0	0
BILL_AMT5	BILL_AMT6	PAY_AMT1	PAY_AMT2	PAY_AMT3	PAY_AMT4	PAY_AMT5	PAY_AMT6
0	0	0	0	0	0	0	0

```
> nrow(data) #500 rows
```

```
[1] 500
```

```
> colnames(data)
```

```
[1] "Y"          "A"          "LIMIT_BAL" "SEX"        "EDUCATION" "AGE"
[7] "PAY_0"      "PAY_2"      "PAY_3"     "PAY_4"      "PAY_5"      "PAY_6"
[13] "BILL_AMT1" "BILL_AMT2" "BILL_AMT3" "BILL_AMT4" "BILL_AMT5" "BILL_AMT6"
[19] "PAY_AMT1"  "PAY_AMT2"  "PAY_AMT3"  "PAY_AMT4"  "PAY_AMT5"  "PAY_AMT6"
```

2 Data manipulation

```
> #factorize cateogorical variables
> data$Y <- as.factor(data$Y)
> data$A <- as.factor(data$A)
> data$SEX <- as.factor(data$SEX)
> data$EDUCATION <- as.factor(data$EDUCATION)
> data$PAY_0 <- as.factor(data$PAY_0)
> data$PAY_2 <- as.factor(data$PAY_2)
> data$PAY_3 <- as.factor(data$PAY_3)
> data$PAY_4 <- as.factor(data$PAY_4)
> data$PAY_5 <- as.factor(data$PAY_5)
> data$PAY_6 <- as.factor(data$PAY_6)
```

```
> #create list of ALL covariates - everything except A and Y
> cov_all <- setdiff(names(data), c("Y", "A"))
> print(cov_all)
```

```
[1] "LIMIT_BAL" "SEX"        "EDUCATION" "AGE"        "PAY_0"      "PAY_2"
[7] "PAY_3"     "PAY_4"     "PAY_5"     "PAY_6"     "BILL_AMT1" "BILL_AMT2"
[13] "BILL_AMT3" "BILL_AMT4" "BILL_AMT5" "BILL_AMT6" "PAY_AMT1"  "PAY_AMT2"
[19] "PAY_AMT3"  "PAY_AMT4"  "PAY_AMT5"  "PAY_AMT6"
```

```
> # Create a version of the data where everyone is enrolled, and no one is enrolled, for outcome regres
> data_all_enrolled <- data
> data_all_enrolled$A <- 1
> data_all_enrolled$A <- as.factor(data_all_enrolled$A)
>
> data_no_enrolled <- data
> data_no_enrolled$A <- 0
> data_no_enrolled$A <- as.factor(data_no_enrolled$A)
> print(unique(data_all_enrolled$A)) #looks good
```

```
[1] 1
Levels: 1
```

```
> print(unique(data_no_enrolled$A))
```

```
[1] 0
Levels: 0
```

3 EDA

4 1 Now the Analysis, for all variables!

5 All Variables: First, untransformed data

6 All Variables, Outcome regression

Basically, model the outcome (Y) of defaulting.

1. Train the model to understand how every variable including PPP enrollment A affects prob of defaulting Y.
2. Take all my customers (A=0 and A=1), and see what their default rate would be if they had A=1.
3. Take all my customers (A=0 and A=1), and see what their default rate would be if they had A=0.
4. ACE is the difference between the two.

```
> formula_or_all_var <- as.formula(paste("Y ~ A +", paste(cov_all, collapse = "+"))) #now A is part of
> model_or_all_var <- glm(formula_or_all_var, data = data, family = binomial) #binomial since 0 or 1 de
>
> print("formula")
```

```
[1] "formula"
```

```
> print(formula_or_all_var)
```

```
Y ~ A + LIMIT_BAL + SEX + EDUCATION + AGE + PAY_0 + PAY_2 + PAY_3 +
    PAY_4 + PAY_5 + PAY_6 + BILL_AMT1 + BILL_AMT2 + BILL_AMT3 +
    BILL_AMT4 + BILL_AMT5 + BILL_AMT6 + PAY_AMT1 + PAY_AMT2 +
    PAY_AMT3 + PAY_AMT4 + PAY_AMT5 + PAY_AMT6
```

```
> # Predict default probability for world of everyone enrolled, and no one enrolled
> data$prob_Y_if_A1_all_var <- predict(model_or_all_var, newdata = data_all_enrolled, type = "response")
> EY_a1_or_all_var <- mean(data$prob_Y_if_A1_all_var)
> data$prob_Y_if_A0_all_var <- predict(model_or_all_var, newdata = data_no_enrolled, type = "response")
> EY_a0_or_all_var <- mean(data$prob_Y_if_A0_all_var)
>
> ace_or_all_var <- EY_a1_or_all_var - EY_a0_or_all_var
>
> #print means
> print("Default likelihood (Full Set, OR), A=1 then A=0")
```

```
[1] "Default likelihood (Full Set, OR), A=1 then A=0"
```

```
> print(EY_a1_or_all_var)
```

```
[1] 0.424858
```

```
> print(EY_a0_or_all_var)
```

```
[1] 0.2184891
```

```
> print(paste("OR ACE (Full Set):", round(ace_or_all_var,4)))
```

```
[1] "OR ACE (Full Set): 0.2064"
```

OR ACE for all vars is 0.2108, meaning enrolling in the PPP is associated with 21% increase in probability of missing a payment. This seems unintuitive...

7 All Variables, IPW

Basically, we know that PPP enrollment is not split evenly between low and high risk customers. We want a way to equalize for underlying riskiness. So we calculate a propensity score to see the likelihood of a customer being enrolled in the PPP. We then look at what ACTUALLY happened (if they were enrolled, and if they charged off). We divide that by the propensity score to ensure that low population outcomes / segments are properly represented/weighted.

1. $P(A=1|L) = P(L|A=1)$: Model the probability a customer is enrolled in the PPP, given their covariates. Basically, use a logistic regression where response variable is A (PPP enrollment), and all variables except A and Y are predictors. Results in function that assigns every customer a probability of being enrolled in PPP - propensity score.
2. $I(A=a)$: indicator flagging to only look at people who actually received the treatment we're interested in. So, if $A=1$, only look at people who were actually enrolled.
3. Y: actual missed payment status, for everyone the indicator flags. Overall: Someone had a 10% chance of enrolling but actually did enroll ($A=1$), and did NOT miss their pmt ($Y=0$): denominator is 0.1, while numerator is 1. So now their missed payment status is 10x the average person in terms of weight.

```
> #get propensity scores
> formula_ps_all_var <- as.formula(paste("A~", paste(cov_all, collapse = "+"))) #formula of A~all covar
> model_ps_all_var <- glm(formula_ps_all_var, data = data, family = binomial) #binomial because only 2
>
> print("formula")
```

```
[1] "formula"
```

```
> print(formula_ps_all_var)
```

```
A ~ LIMIT_BAL + SEX + EDUCATION + AGE + PAY_0 + PAY_2 + PAY_3 +
    PAY_4 + PAY_5 + PAY_6 + BILL_AMT1 + BILL_AMT2 + BILL_AMT3 +
    BILL_AMT4 + BILL_AMT5 + BILL_AMT6 + PAY_AMT1 + PAY_AMT2 +
    PAY_AMT3 + PAY_AMT4 + PAY_AMT5 + PAY_AMT6
```

```
> data$ps_all_var <- predict(model_ps_all_var, type = "response") #calculate propensity scores for all
> print("min and max ps")
```

```
[1] "min and max ps"
```

```
> print(min(data$ps_all_var)) #.199515e-08
```

```
[1] 1.127689e-08
```

```
> print(max(data$ps_all_var)) #1
```

```
[1] 1
```

```
> #extreme ps means likely have extreme scores
>
> data$ps_all_var_clamped <- pmax(pmin(data$ps_all_var, 0.9999), 0.0001) #clamped ps; if 0 or 1 then mo
>
>
> # calculate weights.
> data$weights_all_var <- ifelse(data[["A"]] == 1, 1 / data$ps_all_var_clamped, 1 / (1 - data$ps_all_var
> print("min and max weights")
```

```
[1] "min and max weights"
```

```
> print(min(data$weights_all_var))
```

```
[1] 1.0001
```

```
> print(max(data$weights_all_var))
```

```
[1] 25.74811
```

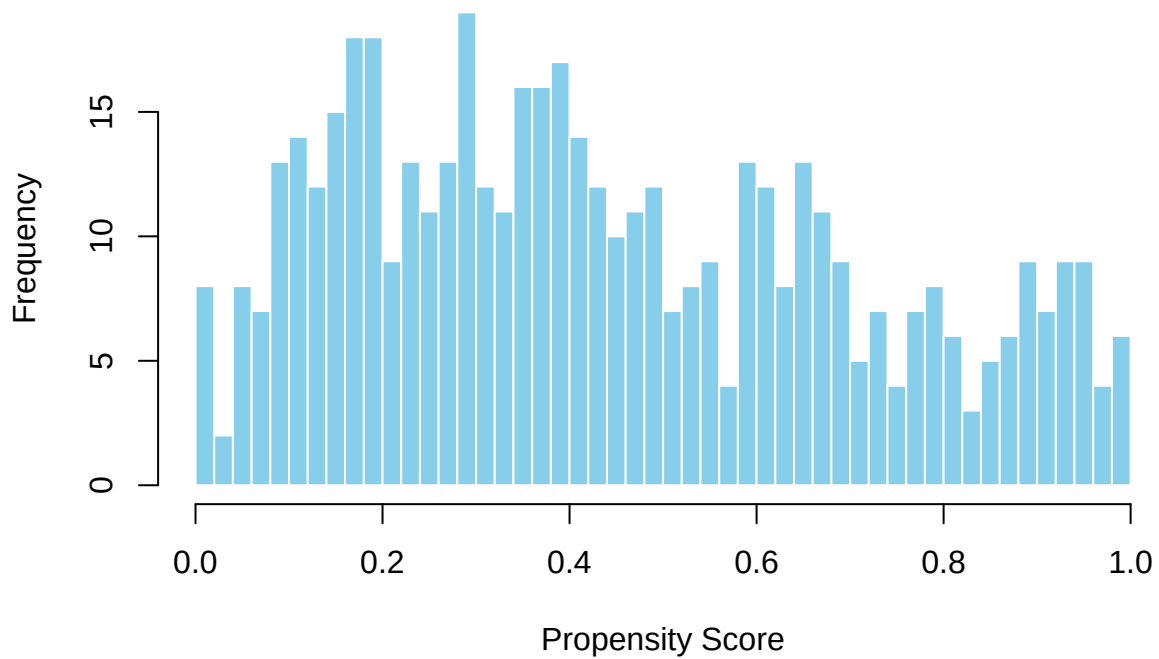
```
> #if customer was actually enrolled (A==1), then d weight as 1/ps, otherwise do 1/(1-ps)  
> #So if customer had A=1 & ps=0.8, then weight = 1.25; but if A=0 then weight=5. Scales by unexpectedn  
>  
> # Estimate ACE using the weighted average  
> #filter to only people who were enrolled A=1 or A=0  
> a0_group <- data[data$A == 0, ]  
> a1_group <- data[data$A == 1, ]  
> # Filter to A0 or A1 groups, get default and multiply by weight; basically, if paid on time, contribu  
> EY_a0_ipw_all_var <- sum(as.numeric(as.character(a0_group$Y)) * a0_group$weights_all_var)/sum(a0_group  
> EY_a1_ipw_all_var <- sum(as.numeric(as.character(a1_group$Y)) * a1_group$weights_all_var)/sum(a1_group  
>  
> ace_ipw_all_var <- EY_a1_ipw_all_var - EY_a0_ipw_all_var  
> print(paste("IPW ACE (Full Set):", round(ace_ipw_all_var,4)))
```

```
[1] "IPW ACE (Full Set): 0.2256"
```

Soem graphs are below just to look at distribution of scores, etc. Given those graphs and the model error from bootstrapping, I had decided to clamp the propensity scores because otherwise we would potentially get a divide by 0 error.

```
> # Create a histogram of Propensity Scores  
> hist(data$ps_all_var,  
+     breaks = 50,  
+     main = "Distribution of Propensity Scores",  
+     xlab = "Propensity Score",  
+     col = "skyblue",  
+     border = "white")
```

Distribution of Propensity Scores



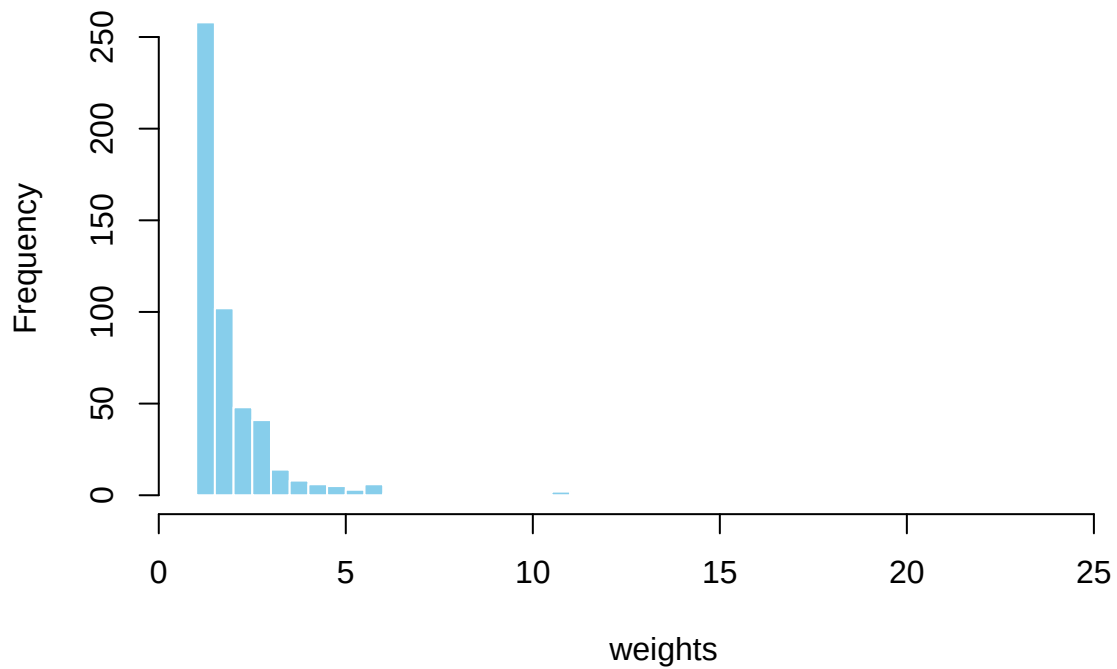
```
> #a lot of low propensity scores

> #Plot by Treatment Group to see "Overlap"
> ggplot(data, aes(x = ps_all_var, fill = as.factor(A))) +
+   geom_density(alpha = 0.5) +
+   labs(title = "Overlap of Propensity Scores",
+         x = "Propensity Score",
+         fill = "Enrolled (A)") +
+   theme_minimal()
```

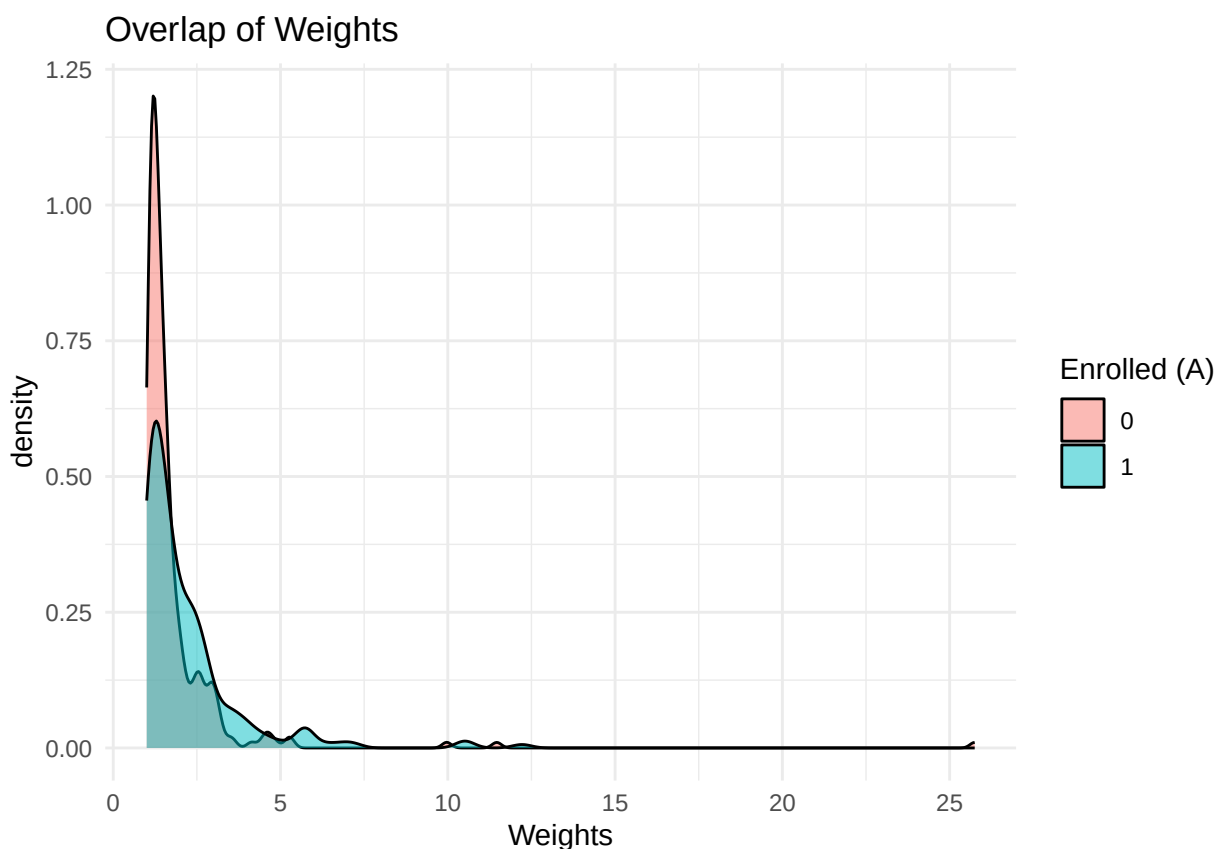


```
> #plot weights  
> hist(data$weights_all_var,  
+       breaks = 50,  
+       main = "Distribution of weights",  
+       xlab = "weights",  
+       col = "skyblue",  
+       border = "white")
```

Distribution of weights



```
> ggplot(data, aes(x = weights_all_var, fill = as.factor(A))) +  
+   geom_density(alpha = 0.5) +  
+   labs(title = "Overlap of Weights",  
+         x = "Weights",  
+         fill = "Enrolled (A)") +  
+   theme_minimal()
```

```
> #no real takeaway. A=1 has higher weights in general maybe?
```

```
> #see how weights are distr A=1 vs A=0
```

```
> data %>%
+   group_by(A) %>%
+   summarise(
+     Count = n(),
+     Min     = min(weights_all_var, na.rm = TRUE),
+     Average = mean(weights_all_var, na.rm = TRUE),
+     Median  = median(weights_all_var, na.rm = TRUE),
+     Perc_25 = quantile(weights_all_var, 0.25, na.rm = TRUE),
+     Perc_75 = quantile(weights_all_var, 0.75, na.rm = TRUE),
+     Max     = max(weights_all_var, na.rm = TRUE),
+     na_count = sum(is.na(weights_all_var)),
+     inf_count = sum(is.infinite(weights_all_var))
+   )
```

```
# A tibble: 2 x 10
```

A	Count	Min	Average	Median	Perc_25	Perc_75	Max	na_count	inf_count
<fct>	<int>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<int>	<int>
1 0	279	1.00	1.79	1.39	1.18	1.80	25.7	0	0
2 1	221	1.00	2.18	1.66	1.22	2.46	12.2	0	0

```
> #although max weight for A=0 is bigger, overall A=1 tends to have higher weights given perc_75, median
```

```
> #no NAs or inf
```

8 All Variables, bootstrap,

Bootstrap to get the bootstrap bias (unsure if important) and variance!

Going forward, I will do bootstrapping so I can capture variance, etc.

Note that I clamp PS scores below, and throughout the rest.

```
> bootstrap_results_all_var <- replicate(bootstrap_sample_size, {  
+   #create bootstrap sample  
+   sample_df <- data[sample(nrow(data), replace = TRUE), ]  
+  
+   #outcome regression  
+   #run model  
+   formula_or_sample_df <- as.formula(paste("Y ~ A +", paste(cov_all, collapse = "+")))  
+   model_or_sample_df <- glm(formula_or_sample_df, data = sample_df, family = binomial)  
+  
+   #create fake datasets  
+   sample_df_a0 <- sample_df  
+   sample_df_a0$A <- factor(0, levels = levels(sample_df$A))  
+   sample_df_a1 <- sample_df  
+   sample_df_a1$A <- factor(1, levels = levels(sample_df$A))  
+  
+   # Predict default probability for world of everyone enrolled, and no one enrolled  
+   sample_df_prob_Y_if_A1 <- predict(model_or_sample_df, newdata = sample_df_a1, type = "response")  
+   EY_a1_or_sample_df <- mean(sample_df_prob_Y_if_A1)  
+  
+   sample_df_prob_Y_if_A0 <- predict(model_or_sample_df, newdata = sample_df_a0, type = "response")  
+   EY_a0_or_sample_df <- mean(sample_df_prob_Y_if_A0)  
+  
+   ace_or_sample_df <- EY_a1_or_sample_df - EY_a0_or_sample_df  
+  
+   #IPW  
+   #get propensity score  
+   formula_ps_sample_df <- as.formula(paste("A ~", paste(cov_all, collapse = "+")))  
+   model_ps_sample_df <- glm(formula_ps_sample_df, data = sample_df, family = binomial)  
+   sample_df$ps_sample_df <- predict(model_ps_sample_df, type = "response")  
+   #clamp ps score between 0.0001 and 0.9999 because otherwise get warnings of fitted prob being 0 or 1  
+   sample_df$ps_sample_df_clamped <- pmax(pmin(sample_df$ps_sample_df, 0.9999), 0.0001)  
+   #get weights  
+   sample_df$weights_sample_df <- ifelse(sample_df$A == 1, 1/sample_df$ps_sample_df_clamped, 1/(1-sample_df$ps_sample_df_clamped))  
+   #divide into a0 and a1 groups  
+   a0_group_sample_df <- sample_df[sample_df$A == 0, ]  
+   a1_group_sample_df <- sample_df[sample_df$A == 1, ]  
+   #calculate ACE  
+   EY_a0_ipw_sample_df <- sum(as.numeric(as.character(a0_group_sample_df$Y))) * a0_group_sample_df$weights_sample_df  
+   EY_a1_ipw_sample_df <- sum(as.numeric(as.character(a1_group_sample_df$Y))) * a1_group_sample_df$weights_sample_df  
+   ace_ipw_sample_df <- EY_a1_ipw_sample_df - EY_a0_ipw_sample_df  
+   return(c(ace_or_sample_df, ace_ipw_sample_df))  
+ })  
  
> # print(bootstrap_results_all_var)
```

Basically, this outputs 2 rows of data ([1,], [2,]). With the 1st row being OR and 2nd being IPW method. and then 10 columns (or however many runs of columns).

```

> orig_ace_all_var <- c(ace_or_all_var, ace_ipw_all_var)
>
> # convert to dataframe
> boot_df_all_var <- as.data.frame(t(bootstrap_results_all_var))
> colnames(boot_df_all_var) <- c("Regression", "IPW")
>
> # Calculate Metrics
> metrics_all_var <- data.frame(
+   Method = c("Regression", "IPW"),
+   Original_ACE = orig_ace_all_var, # Use values from the full dataset
+   Bootstrap_ACE = c(mean(boot_df_all_var$Regression), mean(boot_df_all_var$IPW)),
+   Standard_Error = c(sd(boot_df_all_var$Regression), sd(boot_df_all_var$IPW))
+ )
>
> # Calculate Bias
> metrics_all_var$Bootstrap_Bias <- metrics_all_var$Bootstrap_ACE - metrics_all_var$Original_ACE
> print(metrics_all_var)

```

	Method	Original_ACE	Bootstrap_ACE	Standard_Error	Bootstrap_Bias
1	Regression	0.2063689	0.2146816	0.04388369	0.00831272
2	IPW	0.2255985	0.2362782	0.06317165	0.01067971

This suggests that regression results in a much more stable result, although it is very unintuitive. IPW results in a much higher variance; even with this variance, it is a more intuitive answer as it still suggests a payment plan decreases default risk. Regression is probably biased by overfitting noise and thus creating bias (no proof here, just judgemental unintuitiveness), while IPW could be distorted by extreme weights (as evidenced by high standard error).

9 2 Now the Analysis, for lasso-selected variables!

10 Lasso Variables, Outcome regression , not transformed

```

> #set up the data
> x_or_lasso <- model.matrix(Y ~ A + ., data = data[, c("Y", "A", cov_all)])[-1] #only pull in cov_all
> y_or_lasso <- data$Y
> print(colnames(x_or_lasso))

[1] "A1"          "LIMIT_BAL"  "SEX1"       "EDUCATION2" "EDUCATION3"
[6] "EDUCATION4" "EDUCATION5" "AGE"        "PAY_0-1"    "PAY_00"
[11] "PAY_01"     "PAY_02"     "PAY_03"     "PAY_04"     "PAY_06"
[16] "PAY_2-1"    "PAY_20"     "PAY_22"     "PAY_23"     "PAY_24"
[21] "PAY_25"     "PAY_3-1"    "PAY_30"     "PAY_32"     "PAY_33"
[26] "PAY_34"     "PAY_4-1"    "PAY_40"     "PAY_42"     "PAY_43"
[31] "PAY_5-1"    "PAY_50"     "PAY_52"     "PAY_53"     "PAY_54"
[36] "PAY_6-1"    "PAY_60"     "PAY_62"     "PAY_63"     "BILL_AMT1"
[41] "BILL_AMT2"  "BILL_AMT3"  "BILL_AMT4"  "BILL_AMT5"  "BILL_AMT6"
[46] "PAY_AMT1"   "PAY_AMT2"   "PAY_AMT3"   "PAY_AMT4"   "PAY_AMT5"
[51] "PAY_AMT6"

> col_idx_A1 <- which(colnames(x_or_lasso) == "A1")
> p_fac_or <- rep(1, ncol(x_or_lasso))
> p_fac_or[col_idx_A1] <- 0
>

```

```

> #for A in (penalty factor 0)
> #fit lasso with cross-validation, get best lambda
> model_or_lasso <- cv.glmnet(x_or_lasso, y_or_lasso, family = "binomial", alpha = 1, penalty.factor = 1)
> best_lambda_or_lasso <- model_or_lasso$lambda.min
> print(coef(model_or_lasso))

```

52 x 1 sparse Matrix of class "dgCMatrix"

```

              lambda.1se
(Intercept) -1.36384461
A1           0.91904854
LIMIT_BAL    .
SEX1         .
EDUCATION2   .
EDUCATION3   .
EDUCATION4   .
EDUCATION5   .
AGE          .
PAY_0-1      .
PAY_00       .
PAY_01       .
PAY_02       0.30689616
PAY_03       .
PAY_04       .
PAY_06       .
PAY_2-1      .
PAY_20       .
PAY_22       .
PAY_23       .
PAY_24       .
PAY_25       .
PAY_3-1      .
PAY_30       .
PAY_32       0.78596628
PAY_33       .
PAY_34       .
PAY_4-1      .
PAY_40       .
PAY_42       .
PAY_43       .
PAY_5-1      .
PAY_50       .
PAY_52       .
PAY_53       .
PAY_54       .
PAY_6-1      .
PAY_60       .
PAY_62       0.02140787
PAY_63       .
BILL_AMT1    .
BILL_AMT2    .
BILL_AMT3    .
BILL_AMT4    .
BILL_AMT5    .
BILL_AMT6    .

```

```

PAY_AMT1      .
PAY_AMT2      .
PAY_AMT3      .
PAY_AMT4      .
PAY_AMT5      .
PAY_AMT6      .

> # Predict default probability for world of everyone enrolled, and no one enrolled
>
> #basically, A1 is a dummy column for the factor A=1
> #get the column number for that
> # create new datasets where A is overridden to be all 1s or all 0s, and then transform into model matrix
> x_all_enrolled_or_lasso <- x_or_lasso
> x_all_enrolled_or_lasso[, col_idx_A1] <- 1 # Everyone treated
> x_no_enrolled_or_lasso <- x_or_lasso
> x_no_enrolled_or_lasso[, col_idx_A1] <- 0 # Everyone untreated
>
> #now predict
> data$prob_Y_if_A1_lasso <- as.numeric(predict(model_or_lasso, s = best_lambda_or_lasso, newx = x_all_enrolled_or_lasso))
> EY_a1_or_lasso <- mean(data$prob_Y_if_A1_lasso)
>
> data$prob_Y_if_A0_lasso <- as.numeric(predict(model_or_lasso, s = best_lambda_or_lasso, newx = x_no_enrolled_or_lasso))
> EY_a0_or_lasso <- mean(data$prob_Y_if_A0_lasso)
>
>
> ace_or_lasso <- EY_a1_or_lasso - EY_a0_or_lasso
>
> print("formula")

[1] "formula"

> lasso_y_coefs <- coef(model_or_lasso, s = "lambda.min")
> lasso_y_vars <- rownames(lasso_y_coefs)[which(as.matrix(lasso_y_coefs) != 0)]
> # Remove Intercept and Treatment A to get the adjustment set
> cov_lasso_y <- setdiff(lasso_y_vars, c("(Intercept)", "A"))
> formula_lasso_or_text <- paste("Y ~ ", paste(cov_lasso_y, collapse = " + "))
> formula_lasso_or_text_anotherver <- lasso_y_vars[which(lasso_y_vars != "(Intercept)")]
> print(formula_lasso_or_text)

[1] "Y ~  A1 + LIMIT_BAL + EDUCATION2 + EDUCATION4 + AGE + PAY_0-1 + PAY_00 + PAY_01 + PAY_02 + PAY_03 + PAY_04 + PAY_05 + PAY_06 + PAY_07 + PAY_08 + PAY_09 + PAY_10 + PAY_11 + PAY_12 + PAY_13 + PAY_14 + PAY_15 + PAY_16 + PAY_17 + PAY_18 + PAY_19 + PAY_20 + PAY_21 + PAY_22 + PAY_23 + PAY_24 + PAY_25 + PAY_26 + PAY_27 + PAY_28 + PAY_29 + PAY_30 + PAY_31 + PAY_32 + PAY_33 + PAY_34 + PAY_35 + PAY_36 + PAY_37 + PAY_38 + PAY_39 + PAY_40 + PAY_41 + PAY_42 + PAY_43 + PAY_44 + PAY_45 + PAY_46 + PAY_47 + PAY_48 + PAY_49 + PAY_50 + PAY_51 + PAY_52 + PAY_53 + PAY_54 + PAY_55 + PAY_56 + PAY_57 + PAY_58 + PAY_59 + PAY_60 + PAY_61 + PAY_62 + PAY_63 + PAY_64 + PAY_65 + PAY_66 + PAY_67 + PAY_68 + PAY_69 + PAY_70 + PAY_71 + PAY_72 + PAY_73 + PAY_74 + PAY_75 + PAY_76 + PAY_77 + PAY_78 + PAY_79 + PAY_80 + PAY_81 + PAY_82 + PAY_83 + PAY_84 + PAY_85 + PAY_86 + PAY_87 + PAY_88 + PAY_89 + PAY_90 + PAY_91 + PAY_92 + PAY_93 + PAY_94 + PAY_95 + PAY_96 + PAY_97 + PAY_98 + PAY_99 + PAY_100"

> print(lasso_y_vars)

[1] "(Intercept)" "A1"          "LIMIT_BAL"   "EDUCATION2"  "EDUCATION4"
[6] "AGE"         "PAY_0-1"     "PAY_00"      "PAY_01"      "PAY_02"
[11] "PAY_03"      "PAY_06"     "PAY_24"      "PAY_25"      "PAY_32"
[16] "PAY_34"      "PAY_4-1"    "PAY_42"      "PAY_43"      "PAY_54"
[21] "PAY_62"      "BILL_AMT5"  "PAY_AMT2"    "PAY_AMT3"    "PAY_AMT4"
[26] "PAY_AMT6"

> #print means
> print("Default likelihood (Lasso, OR), A=1 then A=0")

[1] "Default likelihood (Lasso, OR), A=1 then A=0"

> print(EY_a1_or_lasso)

[1] 0.4243885

```

```
> print(EY_a0_or_lasso)
```

```
[1] 0.2211227
```

```
> print(paste("OR ACE (Lasso):", round(ace_or_lasso,4)))
```

```
[1] "OR ACE (Lasso): 0.2033"
```

11 Lasso Variables, IPW

```
> #prepare data
```

```
> x_ps_lasso <- model.matrix(A ~ ., data = data[, c("A", cov_all)])[,-1]
```

```
> y_ps_lasso <- data$A
```

```
>
```

```
>
```

```
> #fit lasso
```

```
> model_ps_lasso <- cv.glmnet(x_ps_lasso, y_ps_lasso, family = "binomial", alpha = 1)
```

```
> best_lambda_ps_lasso <- model_ps_lasso$lambda.min
```

```
>
```

```
>
```

```
> #get propensity scores
```

```
> data$ps_lasso <- as.numeric(predict(model_ps_lasso, s = best_lambda_ps_lasso, newx = x_ps_lasso, type = "link"))
```

```
>
```

```
>
```

```
> print("min and max ps")
```

```
[1] "min and max ps"
```

```
> print(min(data$ps_lasso))
```

```
[1] 0.1381571
```

```
> print(max(data$ps_lasso))
```

```
[1] 0.9413102
```

```
> data$ps_lasso_clamped <- pmax(pmin(data$ps_lasso, 0.9999), 0.0001)
```

```
>
```

```
>
```

```
> # calculate weights.
```

```
> data$weights_lasso <- ifelse(data[["A"]] == 1, 1 / data$ps_lasso_clamped, 1 / (1 - data$ps_lasso_clamped))
```

```
>
```

```
>
```

```
> print("min and max weights")
```

```
[1] "min and max weights"
```

```
> print(min(data$weights_lasso))
```

```
[1] 1.062349
```

```
> print(max(data$weights_lasso))
```

```
[1] 8.964196
```

```
> #get formula
```

```
> print("formula")
```

```
[1] "formula"
```

```
> lasso_a_coefs <- coef(model_ps_lasso, s = "lambda.min")
> lasso_a_vars <- rownames(lasso_a_coefs)[which(as.matrix(lasso_a_coefs) != 0)]
> cov_lasso_a <- setdiff(lasso_a_vars, c("(Intercept)"))
> formula_lasso_ps_text <- paste("A ~", paste(cov_lasso_a, collapse = " + "))
> formula_lasso_ps_text_anotherver <- lasso_a_vars[which(lasso_a_vars != "(Intercept)")]
> print(formula_lasso_ps_text)
```

```
[1] "A ~ SEX1 + EDUCATION3 + EDUCATION5 + AGE + PAY_04 + PAY_AMT6"
```

```
> print(cov_lasso_a)
```

```
[1] "SEX1"          "EDUCATION3" "EDUCATION5" "AGE"          "PAY_04"
[6] "PAY_AMT6"
```

```
> # Estimate ACE using the weighted average
> #filter to only people who were enrolled A=1 or A=0
> a0_group <- data[data$A == 0, ]
> a1_group <- data[data$A == 1, ]
> # Filter to A0 or A1 groups, get default and multiply by weight
> EY_a0_ipw_lasso <- sum(as.numeric(as.character(a0_group$Y)) * a0_group$weights_lasso) / sum(a0_group$weights_lasso)
> EY_a1_ipw_lasso <- sum(as.numeric(as.character(a1_group$Y)) * a1_group$weights_lasso) / sum(a1_group$weights_lasso)
>
>
> ace_ipw_lasso <- EY_a1_ipw_lasso - EY_a0_ipw_lasso
> print(paste("IPW ACE (Lasso):", round(ace_ipw_lasso, 4)))
```

```
[1] "IPW ACE (Lasso): 0.1783"
```

```
> #see how weights are distr A=1 vs A=0
> data %>%
+   group_by(A) %>%
+   summarise(
+     Count = n(),
+     Min = min(weights_lasso, na.rm = TRUE),
+     Average = mean(weights_lasso, na.rm = TRUE),
+     Median = median(weights_lasso, na.rm = TRUE),
+     Perc_25 = quantile(weights_lasso, 0.25, na.rm = TRUE),
+     Perc_75 = quantile(weights_lasso, 0.75, na.rm = TRUE),
+     Max = max(weights_lasso, na.rm = TRUE),
+     na_count = sum(is.na(weights_lasso)),
+     inf_count = sum(is.infinite(weights_lasso))
+   )
```

```
# A tibble: 2 x 10
```

	A	Count	Min	Average	Median	Perc_25	Perc_75	Max	na_count	inf_count
	<fct>	<int>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<int>	<int>
1	0	279	1.16	1.71	1.49	1.33	1.82	8.96	0	0
2	1	221	1.06	2.18	1.90	1.40	2.64	6.03	0	0

```
> #although max weight for A=0 is bigger, overall A=1 tends to have higher weights given perc_75, median
> #no NAs or inf
```

This shows the weights are more reasonable than the full dataset.

12 Lasso Variables, bootstrap

```
> bootstrap_results_lasso <- replicate(bootstrap_sample_size, {
+   #create bootstrap sample
+   sample_df <- data[sample(nrow(data), replace = TRUE), ]
+
+   #outcome regression
+
+   x_or_lasso_sample_df <- model.matrix(Y ~ A + ., data = sample_df[, c("Y", "A", cov_all)])[-1]
+   y_or_lasso_sample_df <- sample_df$Y
+
+   col_idx_A1_sample_df <- which(colnames(x_or_lasso_sample_df) == "A1")
+   p_fac_or_sample_df <- rep(1, ncol(x_or_lasso_sample_df))
+   p_fac_or_sample_df[col_idx_A1_sample_df] <- 0
+
+   #fit lasso with cross-validation, get best lambda
+   model_or_lasso_sample_df <- cv.glmnet(x_or_lasso_sample_df, y_or_lasso_sample_df, family = "binomial")
+   best_lambda_or_lasso_sample_df <- model_or_lasso_sample_df$lambda.min
+
+
+   x_all_enrolled_or_lasso_sample_df <- x_or_lasso_sample_df
+   x_all_enrolled_or_lasso_sample_df[, col_idx_A1_sample_df] <- 1 # Everyone treated
+   x_no_enrolled_or_lasso_sample_df <- x_or_lasso_sample_df
+   x_no_enrolled_or_lasso_sample_df[, col_idx_A1_sample_df] <- 0 # Everyone untreated
+
+   #now predict
+   sample_df_prob_Y_if_A1 <- as.numeric(predict(model_or_lasso_sample_df, s = best_lambda_or_lasso_sample_df, newx = x_all_enrolled_or_lasso_sample_df))
+   EY_a1_or_lasso_sample_df <- mean(sample_df_prob_Y_if_A1)
+
+   sample_df_prob_Y_if_a0 <- as.numeric(predict(model_or_lasso_sample_df, s = best_lambda_or_lasso_sample_df, newx = x_no_enrolled_or_lasso_sample_df))
+   EY_a0_or_lasso_sample_df <- mean(sample_df_prob_Y_if_a0)
+
+   ace_or_lasso_sample_df <- EY_a1_or_lasso_sample_df - EY_a0_or_lasso_sample_df
+
+   #IPW
+
+   #prepare data
+   x_ps_lasso_sample_df <- model.matrix(A ~ ., data = sample_df[, c("A", cov_all)])[-1]
+   y_ps_lasso_sample_df <- sample_df$A
+
+   #fit lasso
+   model_ps_lasso_sample_df <- cv.glmnet(x_ps_lasso_sample_df, y_ps_lasso_sample_df, family = "binomial")
+   best_lambda_ps_lasso_sample_df <- model_ps_lasso_sample_df$lambda.min
+   #get propensity scores
+   sample_df$ps_lasso_sample_df <- as.numeric(predict(model_ps_lasso_sample_df, s = best_lambda_ps_lasso_sample_df, newx = x_ps_lasso_sample_df))
+
+   sample_df$ps_lasso_clamped_sample_df <- pmax(pmin(sample_df$ps_lasso_sample_df, 0.9999), 0.0001)
```



```

+ # calculate weights.
+ sample_df$weights_lasso_sample_df <- ifelse(sample_df[["A"]] == 1, 1 / sample_df$ps_lasso_clamped_sa
+
+ # Estimate ACE using the weighted average
+ #filter to only people who were enrolled A=1 or A=0
+ a0_group_sample_df <- sample_df[sample_df$A == 0, ]
+ a1_group_sample_df <- sample_df[sample_df$A == 1, ]
+ # Filter to A0 or A1 groups, get default and multiply by weight
+ EY_a0_ipw_lasso_sample_df <- sum(as.numeric(as.character(a0_group_sample_df$Y)) * a0_group_sample_df
+ EY_a1_ipw_lasso_sample_df <- sum(as.numeric(as.character(a1_group_sample_df$Y)) * a1_group_sample_df
+
+ ace_ipw_lasso_sample_df <- EY_a1_ipw_lasso_sample_df - EY_a0_ipw_lasso_sample_df
+
+ return(c(ace_or_lasso_sample_df, ace_ipw_lasso_sample_df))
+ })

> orig_ace_lasso <- c(ace_or_lasso, ace_ipw_lasso)
>
>
> # convert to dataframe
> boot_df_lasso <- as.data.frame(t(bootstrap_results_lasso))
> colnames(boot_df_lasso) <- c("Regression", "IPW")
>
>
> # Calculate Metrics
> metrics_lasso <- data.frame(
+ Method = c("Regression", "IPW"),
+ Original_ACE = orig_ace_lasso, # Use values from the full dataset
+ Bootstrap_ACE = c(mean(boot_df_lasso$Regression), mean(boot_df_lasso$IPW)),
+ Standard_Error = c(sd(boot_df_lasso$Regression), sd(boot_df_lasso$IPW))
+ )
>
>
> # Calculate Bias
> metrics_lasso$Bootstrap_Bias <- metrics_lasso$Bootstrap_ACE - metrics_lasso$Original_ACE
> print(metrics_lasso)

```

	Method	Original_ACE	Bootstrap_ACE	Standard_Error	Bootstrap_Bias
1	Regression	0.2032658	0.2044508	0.04012864	0.001185043
2	IPW	0.1782535	0.1952790	0.04284862	0.017025427

This suggests that things are more stable with the lasso.

13 3 Now, do rank PC analysis!!

PC requires data follow Gaussian distribution and have linear relationships - that's why we transformed the variables. The "Rank PC" method aims to make the PC algorithm robust to non-normal distributions and non-linear relationships.

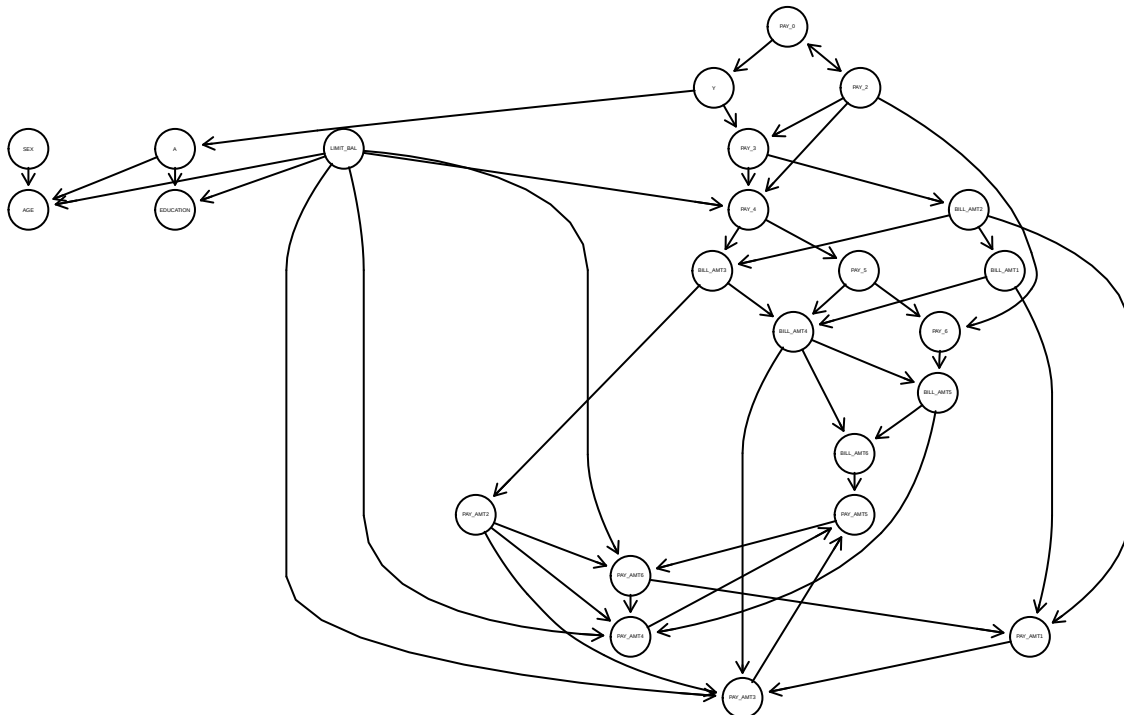
PC systematically tests for conditional independence between variables to determine if there should be an edge - gaussian conditional independence test, assuming variables normally distributed.

Rank PC uses rank based test, say using spearman's rank correlation (considers rank of value not raw values). Does not look at raw values - looks at rank of the value relative to others. Does not assume a specific distribution. Somewhat robust to outliers, because just ranking. On the other hand, if data is not evenly spread, could misrepresent? Because it treats gaps of 100, 101, and 102 as the same as 100, 5k, 100k.

14 Non-transformed, Rank PC

```
> #convert data to numerical for gaussian tests
> rank_pc_data <- data %>%
+   mutate(across(where(is.factor), ~as.numeric(as.factor(.x)))) %>% #rank my factors into numbers
+   select(Y, A, all_of(cov_all))
>
> # calculate correlation matrix and sample size; use to test for conditional independence amongst pairs
> rank_pc_corr_size <- list(C = cor(rank_pc_data, method = "spearman"), n = nrow(rank_pc_data))
>
> # Run PC; alpha serves as a tuning parameter for the significance of independence tests
> rank_pc_fit <- pc(suffStat = rank_pc_corr_size,
+   indepTest = gaussCitest,
+   labels = colnames(rank_pc_data),
+   alpha = 0.10,
+   verbose = FALSE)
>
> # Plot DAG
> plot(rank_pc_fit@graph, main = "Causal DAG: Rank PC Method")
```

Causal DAG: Rank PC Method



```
> #export
> png("OUTPUT_IMAGES_FROM_ANALYSIS_V6_10/causal_dag_rank_pc_method.png", width = 1200, height = 1000, r
>
```

```
> plot(rank_pc_fit@graph, main = "Causal DAG: Rank PC Method")
>
> # 3. Close the device to save the file
> dev.off()
```

pdf

2

```
> rank_pc_dag <- pcalg2dagitty(as(rank_pc_fit@graph, "matrix"),
+                             labels = colnames(rank_pc_data),
+                             type = "cpdag")
>
> # Get the Markov Blanket (Parents + Children + Spouses)
> rank_pc_mb_y <- markovBlanket(rank_pc_dag, "Y")
> print(rank_pc_mb_y)
```

```
[1] "A"      "PAY_3" "PAY_0"
```

```
> # Update your covariate list for the models
> cov_rank_pc_y <- setdiff(rank_pc_mb_y, "A") #exclude A
> print(cov_rank_pc_y) #only PAY_0?
```

```
[1] "PAY_3" "PAY_0"
```

```
> # Get the Markov Blanket (Parents + Children + Spouses)
> rank_pc_mb_a <- markovBlanket(rank_pc_dag, "A")
> print(rank_pc_mb_a)
```

```
[1] "AGE"      "EDUCATION" "Y"      "PAY_3"
```

```
> # Update your covariate list for the models
> cov_rank_pc_a <- setdiff(rank_pc_mb_a, "Y") #exclude A
> print(cov_rank_pc_a) #only PAY_0?
```

```
[1] "AGE"      "EDUCATION" "PAY_3"
```

rank pc algorithm nontransformed or

```
``` r
> #outcome regression
> if(length(cov_rank_pc_y) == 0) {
+ formula_or_rank_pc <- as.formula("Y ~ A")
+ } else {
+ formula_or_rank_pc <- as.formula(paste("Y ~ A +", paste(cov_rank_pc_y, collapse = "+")))
+ }
> model_or_rank_pc <- glm(formula_or_rank_pc, data = data, family = binomial)
>
> print("formula")
[1] "formula"
> print(formula_or_rank_pc)
```

```
Y ~ A + PAY_3 + PAY_0
```

```
> # pc_data_all_enrolled <- data_all_enrolled %>%
> # mutate(across(everything(), as.numeric)) %>%
```

```

> # select(Y, A, all_of(cov_all))
>
> # pc_data_no_enrolled <- data_no_enrolled %>%
> # # mutate(across(everything(), as.numeric)) %>%
> # select(Y, A, all_of(cov_all))
>
> # Predict default probability for world of everyone enrolled, and no one enrolled
> data$prob_Y_if_A1_rank_pc_var<- predict(model_or_rank_pc, newdata = data_all_enrolled, type = "response")
>
> EY_a1_or_rank_pc_var <- mean(data$prob_Y_if_A1_rank_pc_var)
>
> data$prob_Y_if_A0_rank_pc_var <- predict(model_or_rank_pc, newdata = data_no_enrolled, type = "response")
> EY_a0_or_rank_pc_var <- mean(data$prob_Y_if_A0_rank_pc_var)
>
> ace_or_rank_pc_var <- EY_a1_or_rank_pc_var - EY_a0_or_rank_pc_var
>
> #print means
> print("Default likelihood (Rank PC Set, OR), A=1 then A=0")

[1] "Default likelihood (Rank PC Set, OR), A=1 then A=0"
> print(EY_a1_or_rank_pc_var)

[1] 0.4266281
> print(EY_a0_or_rank_pc_var)

[1] 0.2202481
> print(paste("OR ACE (Rank PC Set):", round(ace_or_rank_pc_var,4)))

[1] "OR ACE (Rank PC Set): 0.2064"

```

## 15 Rank PC algorithm nonttransformed ipw

```

> if(length(cov_rank_pc_a) == 0) {
+ formula_ps_rank_pc_var <- as.formula("A ~ 1")
+ } else {
+ formula_ps_rank_pc_var <- as.formula(paste("A~", paste(cov_rank_pc_a, collapse = "+"))) #formula of
+ }
> #get propensity scores
> model_ps_rank_pc_var <- glm(formula_ps_rank_pc_var, data = data, family = binomial) #binomial because
>
> print("formula")

[1] "formula"
> print(formula_ps_rank_pc_var)

A ~ AGE + EDUCATION + PAY_3
> data$ps_rank_pc_var <- predict(model_ps_rank_pc_var, type = "response") #calculate propensity scores
> print("min and max ps")

[1] "min and max ps"

```

```

> print(min(data$ps_rank_pc_var))

[1] 1.28331e-06
> print(max(data$ps_rank_pc_var))

[1] 0.9746681
> #extreme ps means likely have extreme scores
>
> data$ps_rank_pc_var_clamped <- pmax(pmin(data$ps_rank_pc_var, 0.9999), 0.0001)
>
>
> # calculate weights.
> data$weights_rank_pc_var <- ifelse(data[["A"]] == 1, 1 / data$ps_rank_pc_var_clamped, 1 / (1 - data$ps_rank_pc_var_clamped))
>
> print("min and max weights")

[1] "min and max weights"
> print(min(data$weights_rank_pc_var)) #1.0001

[1] 1.0001
> print(max(data$weights_rank_pc_var)) #2.8

[1] 14.71795
> #if customer was actually enrolled (A==1), then d weight as 1/ps, otherwise do 1/(1-ps)
> #So if customer had A=1 & ps=0.8, then weight = 1.25; but if A=0 then weight=5. Scales by unexpected
>
> # Estimate ACE using the weighted average
> #filter to only people who were enrolled A=1 or A=0
> a0_group <- data[data$A == 0,]
> a1_group <- data[data$A == 1,]
> # Filter to A0 or A1 groups, get default and multiply by weight; basically, if paid on time, contribu
> EY_a0_ipw_rank_pc_var <- sum(as.numeric(as.character(a0_group$Y)) * a0_group$weights_rank_pc_var)/sum(a0_group$weights_rank_pc_var)
> EY_a1_ipw_rank_pc_var <- sum(as.numeric(as.character(a1_group$Y)) * a1_group$weights_rank_pc_var)/sum(a1_group$weights_rank_pc_var)
>
> ace_ipw_rank_pc_var <- EY_a1_ipw_rank_pc_var - EY_a0_ipw_rank_pc_var
> print(paste("IPW ACE (Rank PC set):", round(ace_ipw_rank_pc_var,4)))

[1] "IPW ACE (Rank PC set): 0.1827"

```

## 16 Rank Pc algorithm bootstrap nontransformed

```

> bootstrap_results_rank_pc <- replicate(bootstrap_sample_size, {
+ #bootstrap sample
+ sample_df <- data[sample(nrow(data), replace = TRUE),]
+
+ #create pc markov blanket
+ boot_data <- sample_df %>%
+ mutate(across(where(is.factor), ~as.numeric(as.factor(.x)))) %>% #rank my factors into numbers
+ select(Y, A, all_of(cov_all))
+
+ #get correlation, sample size of bootstrapped data

```

```

+ suff_boot <- list(C = cor(boot_data, method = "spearman"), n = nrow(boot_data))
+
+ #Run rank PC Algorithm
+ fit_boot <- pc(suffStat = suff_boot, indepTest = gaussCitest,
+ labels = colnames(boot_data), alpha = 0.10, verbose = FALSE)
+
+ adj_matrix <- as(fit_boot@graph, "matrix")
+ adj_matrix_input <- pcalg2dagitty(adj_matrix,
+ labels = colnames(boot_data),
+ type = "cpdag")
+
+ # Get the Markov Blanket (Parents + Children + Spouses)
+ y_indices_all_boot <- markovBlanket(adj_matrix_input, "Y")
+ cov_boot <- setdiff(y_indices_all_boot, "A") #exclude A
+
+ a_indices_all_boot <- markovBlanket(adj_matrix_input, "A")
+ cov_boot_a <- setdiff(a_indices_all_boot, "Y")
+
+ #what is markov blanket empty?
+ if(length(cov_boot) == 0) {
+ formula_or_boot <- as.formula("Y ~ A")
+ } else {
+ formula_or_boot <- as.formula(paste("Y ~ A +", paste(cov_boot, collapse = "+")))
+ }
+ if(length(cov_boot_a) == 0) {
+ formula_ps_boot <- as.formula("A ~ 1")
+ } else {
+ formula_ps_boot <- as.formula(paste("A ~", paste(cov_boot_a, collapse = "+")))
+ }
+
+ # outcome regression
+ model_or_boot <- glm(formula_or_boot, data = sample_df, family = binomial)
+
+ #create fake datasets
+ sample_df_a0 <- sample_df
+ sample_df_a0$A <- factor(0, levels = levels(sample_df$A))
+ sample_df_a1 <- sample_df
+ sample_df_a1$A <- factor(1, levels = levels(sample_df$A))
+
+ # Predict default probability for world of everyone enrolled, and no one enrolled
+ sample_df_prob_Y_if_A1 <- predict(model_or_boot, newdata = sample_df_a1, type = "response")
+ EY_a1_or_sample_df <- mean(sample_df_prob_Y_if_A1)
+
+ sample_df_prob_Y_if_A0 <- predict(model_or_boot, newdata = sample_df_a0, type = "response")
+ EY_a0_or_sample_df <- mean(sample_df_prob_Y_if_A0)
+
+ ace_or_sample_df <- EY_a1_or_sample_df - EY_a0_or_sample_df
+
+ #IPW
+ #get propensity score
+ model_ps_sample_df <- glm(formula_ps_boot, data = sample_df, family = binomial)
+ sample_df$ps_sample_df <- predict(model_ps_sample_df, type = "response")
+ #clamp ps score between 0.0001 and 0.9999 because otherwise get warnings of fitted prob being 0 or

```

```

+ sample_df$ps_sample_df_clamped <- pmax(pmin(sample_df$ps_sample_df, 0.9999), 0.0001)
+ #get weights
+ sample_df$weights_sample_df <- ifelse(sample_df$A == 1, 1/sample_df$ps_sample_df_clamped, 1/(1-sample_df$ps_sample_df_clamped))
+ #divide into a0 and a1 groups
+ a0_group_sample_df <- sample_df[sample_df$A == 0,]
+ a1_group_sample_df <- sample_df[sample_df$A == 1,]
+ #calculate ACE
+ EY_a0_ipw_sample_df <- sum(as.numeric(as.character(a0_group_sample_df$Y)) * a0_group_sample_df$weights_sample_df)
+ EY_a1_ipw_sample_df <- sum(as.numeric(as.character(a1_group_sample_df$Y)) * a1_group_sample_df$weights_sample_df)
+ ace_ipw_sample_df <- EY_a1_ipw_sample_df - EY_a0_ipw_sample_df
+
+ return(list(ace_or = ace_or_sample_df,
+ ace_ipw = ace_ipw_sample_df,
+ adj_matrix = adj_matrix,
+ blanket = y_indices_all_boot,
+ blanket_a = a_indices_all_boot
+))
+ }, simplify = FALSE)

> orig_ace_rank_pc_var <- c(ace_or_rank_pc_var, ace_ipw_rank_pc_var)
>
> #extract ACE values
> boot_aces_or_rank_pc <- unlist(lapply(bootstrap_results_rank_pc, function(x) x$ace_or))
> boot_aces_ipw_rank_pc <- unlist(lapply(bootstrap_results_rank_pc, function(x) x$ace_ipw))
>
> #extract markov blanket
> boot_blankets_rank_pc <- lapply(bootstrap_results_rank_pc, function(x) x$blanket)
> boot_blankets_rank_pc_a <- lapply(bootstrap_results_rank_pc, function(x) x$blanket_a)
>
> # convert to dataframe
> boot_df_rank_pc_var <- data.frame(
+ Regression = boot_aces_or_rank_pc,
+ IPW = boot_aces_ipw_rank_pc
+)
> colnames(boot_df_rank_pc_var) <- c("Regression", "IPW")
>
> # Calculate Metrics
> metrics_rank_pc_var <- data.frame(
+ Method = c("Regression", "IPW"),
+ Original_ACE = orig_ace_rank_pc_var, # Use values from the full dataset
+ Bootstrap_ACE = c(mean(boot_df_rank_pc_var$Regression), mean(boot_df_rank_pc_var$IPW)),
+ Standard_Error = c(sd(boot_df_rank_pc_var$Regression), sd(boot_df_rank_pc_var$IPW))
+)
>
> # Calculate Bias
> metrics_rank_pc_var$Bootstrap_Bias <- metrics_rank_pc_var$Bootstrap_ACE - metrics_rank_pc_var$Original_ACE
> print(metrics_rank_pc_var)

 Method Original_ACE Bootstrap_ACE Standard_Error Bootstrap_Bias
1 Regression 0.2063799 0.2029024 0.04136650 -0.003477475
2 IPW 0.1826597 0.1880597 0.04670383 0.005399950

> boot_blankets_rank_pc_unlisted <- unlist(boot_blankets_rank_pc)
>

```

```
> boot_mb_stability_rank_pc <- table(boot_blankets_rank_pc_unlisted) / bootstrap_sample_size
> print("Selection Frequency for Y's Markov Blanket:")
```

```
[1] "Selection Frequency for Y's Markov Blanket:"
```

```
> print(sort(boot_mb_stability_rank_pc, decreasing = TRUE))
```

```
boot_blankets_rank_pc_unlisted
```

	A	PAY_0	PAY_3	PAY_2	PAY_AMT2	PAY_5	PAY_6	AGE
	0.988	0.981	0.572	0.336	0.276	0.219	0.191	0.175
PAY_AMT4	PAY_AMT3	PAY_AMT1	EDUCATION	PAY_4	LIMIT_BAL	SEX	PAY_AMT5	
0.109	0.087	0.062	0.060	0.049	0.029	0.028	0.005	
BILL_AMT2	PAY_AMT6	BILL_AMT1						
0.002	0.002	0.001						

```
> boot_blankets_rank_pc_unlisted_a <- unlist(boot_blankets_rank_pc_a)
```

```
>
```

```
> boot_mb_stability_rank_pc_a <- table(boot_blankets_rank_pc_unlisted_a) / bootstrap_sample_size
> print("Selection Frequency for A's Markov Blanket:")
```

```
[1] "Selection Frequency for A's Markov Blanket:"
```

```
> print(sort(boot_mb_stability_rank_pc_a, decreasing = TRUE))
```

```
boot_blankets_rank_pc_unlisted_a
```

	Y	AGE	EDUCATION	PAY_3	PAY_0	PAY_AMT2	LIMIT_BAL	PAY_AMT4
	0.988	0.986	0.744	0.383	0.280	0.235	0.191	0.171
SEX	PAY_AMT3	PAY_AMT6	PAY_AMT1	PAY_AMT5	BILL_AMT1	PAY_5	PAY_4	
0.129	0.097	0.053	0.035	0.014	0.007	0.006	0.005	
PAY_2	BILL_AMT4	PAY_6	BILL_AMT2	BILL_AMT6				
0.004	0.002	0.002	0.001	0.001				

```
> #get adjacent matrices
```

```
> all_adj_bootstrap_rank_pc <- lapply(bootstrap_results_rank_pc, function(x) x$adj_matrix)
```

```
> #sum them up to get edge frequencies - prob an edge exists between 2 variables
```

```
> consensus_matrix_bootstrap_rank_pc <- Reduce("+", all_adj_bootstrap_rank_pc) / bootstrap_sample_size
```

```
>
```

```
> # Create a graph where an edge exists if it appeared in > 50% of samples
```

```
> average_adj_bootstrap_rank_pc <- (consensus_matrix_bootstrap_rank_pc > 0.5) * 1
```

```
>
```

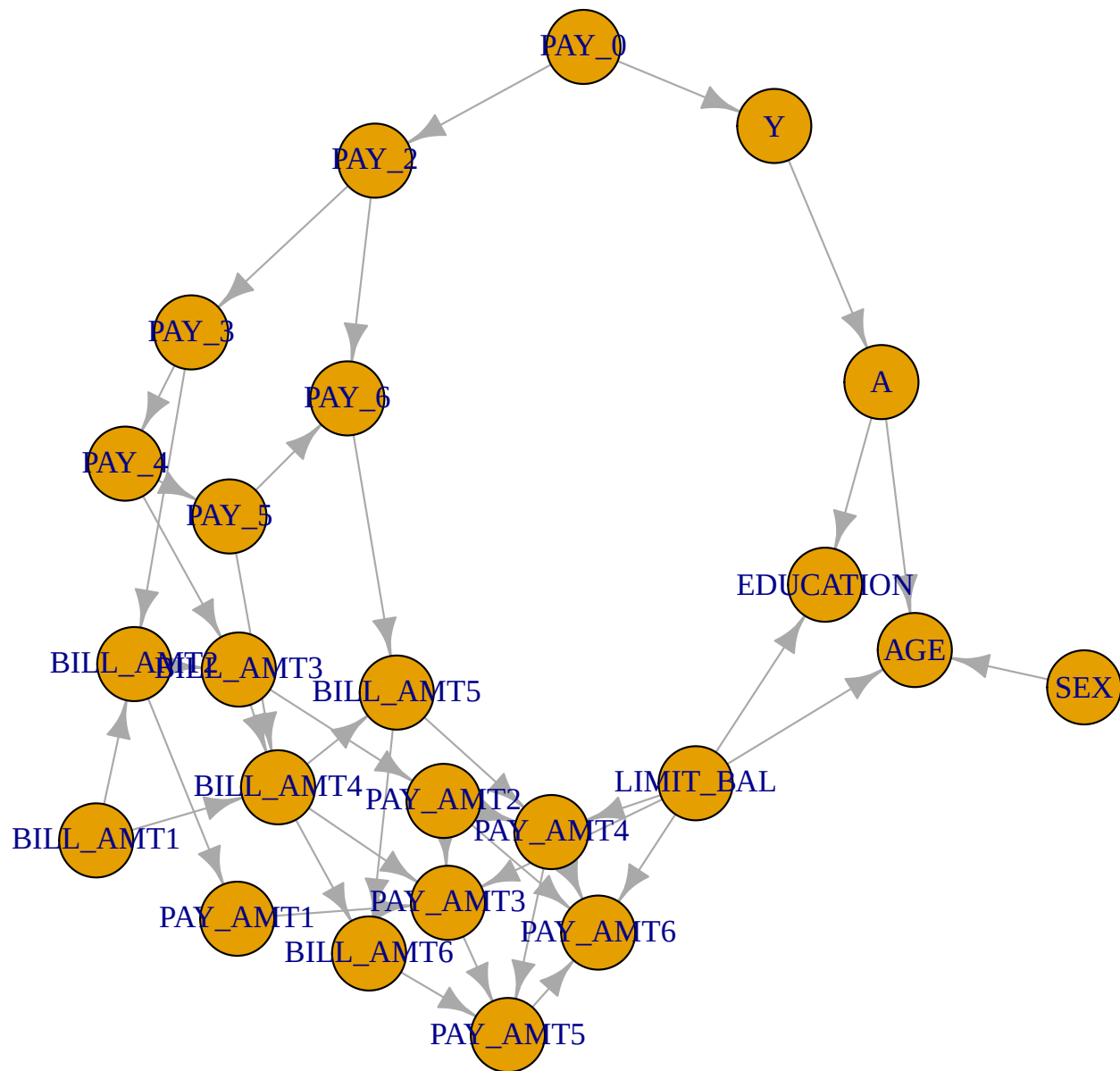
```
> # Convert back to a graph object for plotting
```

```
> avg_graph_bootstrap_rank_pc <- graph_from_adjacency_matrix(average_adj_bootstrap_rank_pc, mode = "dir
```

```
> plot(avg_graph_bootstrap_rank_pc, main = "Consensus DAG (Edges in >50% of Bootstraps)")
```



## Consensus DAG (Edges in >50% of Bootstraps)



```

> png("OUTPUT_IMAGES_FROM_ANALYSIS_V6_10/consensus_dag_rank_pc_bootstrap.png", width = 1200, height = 1000)
> par(mar = c(1, 1, 3, 1))
> plot(avg_graph_bootstrap_rank_pc,
+ main = "Consensus DAG (Edges in >50% of Bootstraps): Rank PC Method",
+ vertex.size = 15, # Smaller nodes
+ vertex.label.cex = 0.6, # Smaller labels
+ vertex.label.dist = 1.2, # Offset labels from node centers
+ edge.arrow.size = 0.3, # Smaller arrows
+ edge.curved = 0.2) # Slight curve to distinguish overlapping arrows
> dev.off()

```

pdf

## 17 4 Now the Analysis, for PC-selected variables! (limited vars to most recent month first)

## 18 PC Algorithm

Find the markov blanket of Y, consisting of its parents, children, and parents of its children. Y is conditionally independent of all other variables GIVEN its Markov Blanket.

So: use PC algorithm to find the full DAG. Identify variables directly connected to Y. Run OR and IPW using the variables directly connected to Y.

Note: assumes 1. Causal sufficiency - no hidden confounders 2. faithfulness: conditional independencies in the data imply cond ind in the DAG. 3. Markov condition: variable is conditionally independent of its non-descendents given its parents, in the DAG.

PC measures cond ind in the data, and searches for a graph to satisfy them.

1. start with fully connected, UNDIRECTED graph
2. test for conditional independence amongst all pairs nad subsets of variables  $\rightarrow$  X ind of Y given Z, for different sets of Z
3. remove edges based on these tests
4. continue until we can't remove more edges without violating conditional independences in data
5. apply orientation rules (colliders etc) to orient undirected edges, as possible
6. adjust to make sure graph remains acyclic

For Gaussian data, this is like testing if the null hypothesis is rho of X&Y given Z is 0, vs not. If data is categorical, can test conditional independence using a chi-square test.

Obtain the DAG.

The “Penalty” of Categorical Data

When you use a Gaussian test (like gaussCIttest), the algorithm treats everything as a simple linear correlation. It only needs to find one “slope” to connect two variables.

mixCIttest is different. When it tests a factor (like EDUCATION with 4 levels) against your outcome Y: It essentially creates a contingency table or a grouped model. If you only have 500 rows, the data gets split into very small “buckets” (e.g., how many people with “Graduate School” education actually “Defaulted”?). If those buckets are too small, the p-value becomes very large (insignificant), and the algorithm deletes the line. 2. Loss of Power in Conditional Tests The PC algorithm doesn't just test A vs B. It tests A vs B given C, D, and E. As the “Conditioning Set” grows, mixCIttest has to slice your 500 rows even thinner. It is much harder to prove that LIMIT\_BAL and Y are dependent after accounting for PAY\_0 and EDUCATION when using a mixed-type test. The test “gives up” more easily than a purely numeric one would. 3. “Conservative” by Design The micd package (which provides mixCIttest) is designed to avoid False Positives. It would rather tell you two variables are independent than show a “fake” causal link. In biostatistics and high-stakes financial modeling, a “Sparse” graph is often considered more trustworthy than a “Spaghetti” graph where everything is connected to everything.

```
> pc_data <- data %>%
+ mutate(across(where(is.integer), as.numeric)) %>% #convert all integers to numeric, to run
+ select(Y, A, all_of(cov_all))
> # pc_data_var_types <- ifelse(sapply(pc_data, is.factor), 1, 0)
> print(summary(pc_data)) #proves data is in right format.
```

Y	A	LIMIT_BAL	SEX	EDUCATION	AGE
0:344	0:279	Min. : 10000	0:291	1:176	Min. :21.00

1:156	1:221	1st Qu.: 50000	1:209	2:241	1st Qu.:28.00
		Median :140000		3: 74	Median :34.00
		Mean :164060		4: 2	Mean :35.35
		3rd Qu.:230000		5: 7	3rd Qu.:40.00
		Max. :620000			Max. :65.00

	PAY_0	PAY_2	PAY_3	PAY_4	PAY_5	PAY_6	BILL_AMT1
0	:232	-2: 64	-2: 68	-2: 74	-2: 72	-2: 76	Min. : -532
-1	:100	-1:109	-1:115	-1:106	-1: 99	-1:102	1st Qu.: 3744
1	: 71	0 :255	0 :255	0 :248	0 :270	0 :265	Median : 22156
-2	: 48	2 : 63	2 : 57	2 : 69	2 : 56	2 : 54	Mean : 49072
2	: 42	3 : 5	3 : 4	3 : 3	3 : 2	3 : 3	3rd Qu.: 58694
3	: 4	4 : 3	4 : 1		4 : 1		Max. :604019

(Other): 3 5 : 1

BILL_AMT2	BILL_AMT3	BILL_AMT4	BILL_AMT5
Min. : -17710	Min. : -17706	Min. : -2800	Min. : -81334
1st Qu.: 2418	1st Qu.: 2583	1st Qu.: 2888	1st Qu.: 2451
Median : 19870	Median : 19507	Median : 18654	Median : 18538
Mean : 45531	Mean : 42682	Mean : 40983	Mean : 38318
3rd Qu.: 55343	3rd Qu.: 51370	3rd Qu.: 49070	3rd Qu.: 45385
Max. :605943	Max. :467911	Max. :404157	Max. :587067

BILL_AMT6	PAY_AMT1	PAY_AMT2	PAY_AMT3
Min. : -2506	Min. : 0	Min. : 0	Min. : 0.0
1st Qu.: 1228	1st Qu.: 1000	1st Qu.: 1000	1st Qu.: 498.2
Median : 17972	Median : 2000	Median : 2038	Median : 1737.5
Mean : 36065	Mean : 4940	Mean : 5766	Mean : 5872.4
3rd Qu.: 44432	3rd Qu.: 4753	3rd Qu.: 5000	3rd Qu.: 4906.8
Max. :323821	Max. :234000	Max. :294318	Max. :319494.0

PAY_AMT4	PAY_AMT5	PAY_AMT6
Min. : 0	Min. : 0	Min. : 0.0
1st Qu.: 298	1st Qu.: 390	1st Qu.: 181.5
Median : 1576	Median : 1529	Median : 1500.0
Mean : 4258	Mean : 5212	Mean : 5084.9
3rd Qu.: 3956	3rd Qu.: 4000	3rd Qu.: 4134.0
Max. :109332	Max. :331788	Max. :189600.0

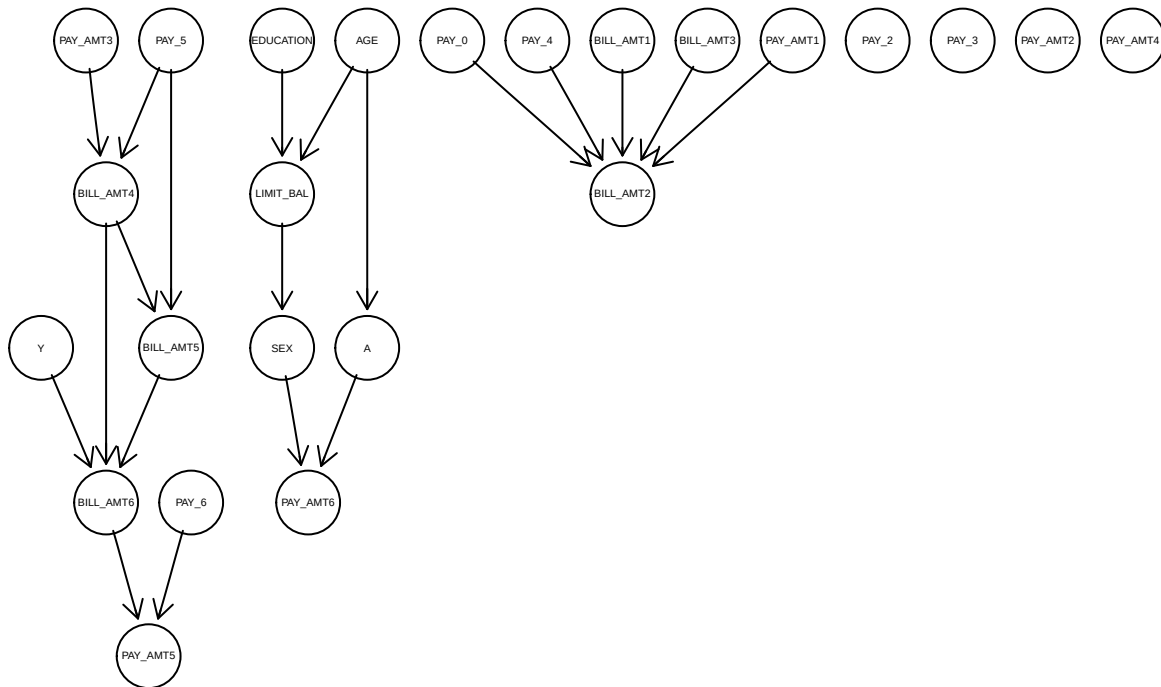
```

> # attr(pc_data, "adaptDF") <- TRUE
>
> # Run PC; alpha serves as a tuning parameter for the significance of independence tests
> pc_fit_test <- pc(suffStat = pc_data,
+ indepTest = mixCitest,
+ labels = colnames(pc_data),
+ alpha = 0.10,
+ verbose = FALSE,
+ u2pd = "relaxed",
+ skel.method = "stable" , #order of test doesnt matter
+ maj.rule = FALSE, #if some say a-->b and some say b-->a, then pick direction with
+ solve.confl = FALSE #if some say a-->b and some say b-->a, then pick a direction
+)
>
> # Plot DAG

```

```
> plot(pc_fit_test@graph, main = "TEST Causal DAG: Markov Blanket Discovery (PC Method)")
```

## TEST Causal DAG: Markov Blanket Discovery (PC Method)



```

> #export
> png("OUTPUT_IMAGES_FROM_ANALYSIS_V6_10/causal_dag_pc_method_test.png", width = 1200, height = 1000, res = 300)
>
> plot(pc_fit_test@graph, main = "TEST Causal DAG: Markov Blanket Discovery (PC Method)")
>
> # 3. Close the device to save the file
> dev.off()

```

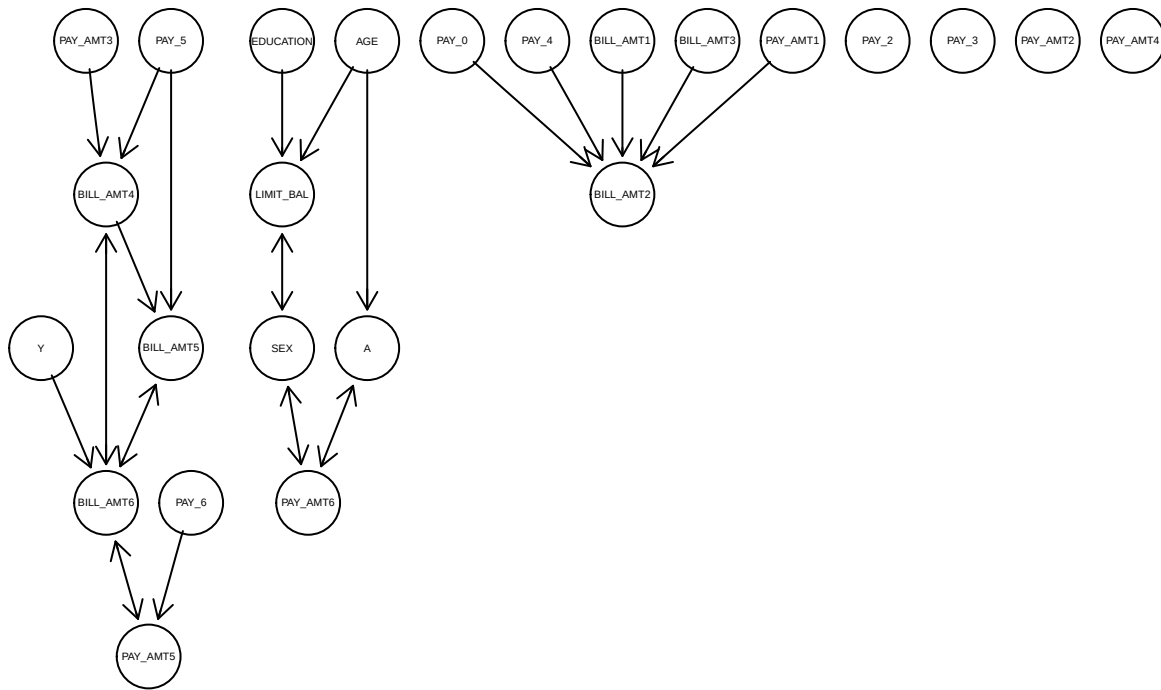
pdf  
2

```

> # Run PC; alpha serves as a tuning parameter for the significance of independence tests
> pc_fit_test <- pc(suffStat = pc_data,
+ indepTest = mixCItest,
+ labels = colnames(pc_data),
+ alpha = 0.10,
+ verbose = FALSE,
+ u2pd = "relaxed",
+ skel.method = "stable", #order of test doesnt matter
+ maj.rule = FALSE,
+ solve.confl = TRUE #if some say a-->b and some say b-->a, then pick a direction
+)
>
> # Plot DAG
> plot(pc_fit_test@graph, main = "TEST Causal DAG: Markov Blanket Discovery (PC Method)")

```

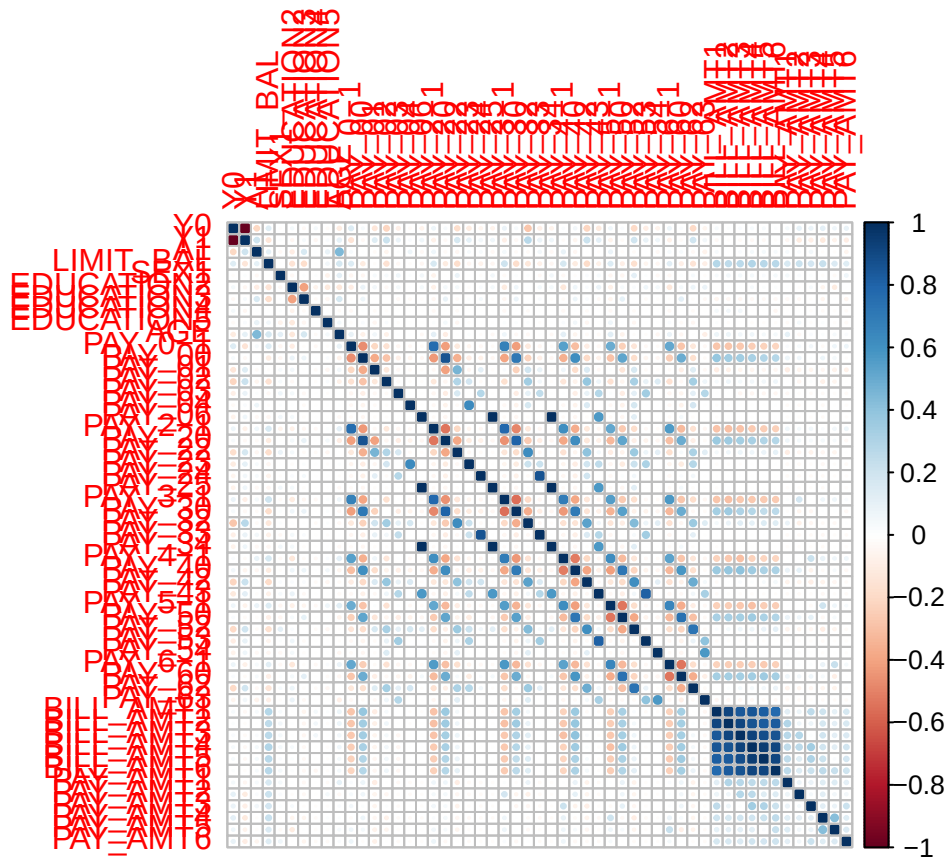
## TEST Causal DAG: Markov Blanket Discovery (PC Method)



```
> #export
> png("OUTPUT_IMAGES_FROM_ANALYSIS_V6_10/causal_dag_pc_method_v2_test.png", width = 1200, height = 1000)
>
> plot(pc_fit_test@graph, main = "TEST Causal DAG: Markov Blanket Discovery (PC Method)")
>
> # 3. Close the device to save the file
> dev.off()
```

pdf  
2

```
> corrplot(cor(model.matrix(~.-1, data=pc_data)), method="circle")
```



```
> png("OUTPUT_IMAGES_FROM_ANALYSIS_V6_10/correlation_matrix_default_test.png", width = 1000, height = 1000)
> corrplot(cor(model.matrix(~.-1, data=pc_data)), method="circle")
> dev.off()
```

pdf  
2

Note the correlation is really low, resulting in the PC algorithm not finding many edges. This is likely because of the categorical variables- the mixCIttest is struggling to find significant relationships because of the small effective sample size when conditioning on multiple variables. This is not solved even when we limit to max conditioning set size of 2.

```
> #convert data to numerical for gaussian tests
> pc_data <- data %>%
+ mutate(across(where(is.integer), as.numeric)) %>% #convert all integers to numeric, to run
+ select(Y, A, all_of(cov_all))
>
> # calculate correlation matrix and sample size; use to test for conditional independence amongst pairs
> # pc_corr_size <- list(C = cor(pc_data), n = nrow(pc_data))
>
>
> pc_fit_bnlearn <- pc.stable(pc_data,
+ # blacklist = master_blacklist,
+ # test = "mix", #not needed, will automatically detect which test based on variable types
+ alpha = 0.10)
>
>
> # Get adjacency matrix from PC fit
```

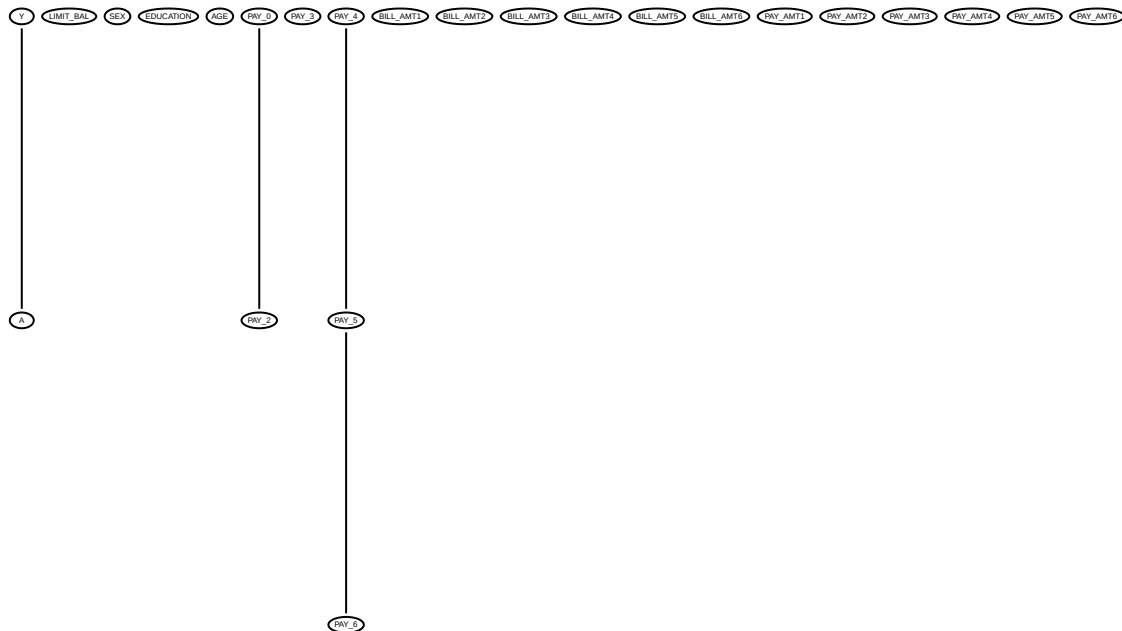
```

> # pc_fit_adj_matrix <- amat(pc_fit)
> # print(pc_fit_adj_matrix)
>
> pc_fit_bnlearn_cextend <- cextend(pc_fit_bnlearn) #directs edges as much as possible, but doesn't ori
>
>
> pc_fit_bnlearn_maxsx <- pc.stable(pc_data,
+ # blacklist = master_blacklist,
+ # test = "mix", #not needed, will automatically detect which test based on vari
+ alpha = 0.10,
+ max.sx=2) #limits size of conditioning set to 2, which should help with the power
> pc_fit_bnlearn_maxsx_cextend <- cextend(pc_fit_bnlearn_maxsx) #directs edges as much as possible, but
>
> cov_most_recent <- c("PAY_0", "EDUCATION", "AGE", "SEX", "PAY_AMT1", "BILL_AMT1", "LIMIT_BAL")
>
> pc_data_limitedcov <- pc_data %>%
+ select(Y, A, all_of(cov_most_recent))
> pc_fit_bnlearn_limitedcov <- pc.stable(pc_data_limitedcov,
+ # blacklist = master_blacklist,
+ # test = "mix", #not needed, will automatically detect which test based on vari
+ alpha = 0.10)

> # Plot DAG
> graphviz.plot(pc_fit_bnlearn, shape = "ellipse",
+ main = "TEST Full Causal DAG, PC Method")

```

## TEST Full Causal DAG, PC Method



```

> # 2. Export a high-quality version
> png("OUTPUT_IMAGES_FROM_ANALYSIS_V6_10/causal_dag_pc_method_recentmonth_test.png", width = 1600, height = 1000)
> graphviz.plot(pc_fit_bnlearn, shape = "ellipse",
+ main = "TEST Full Causal DAG, PC Method")

```

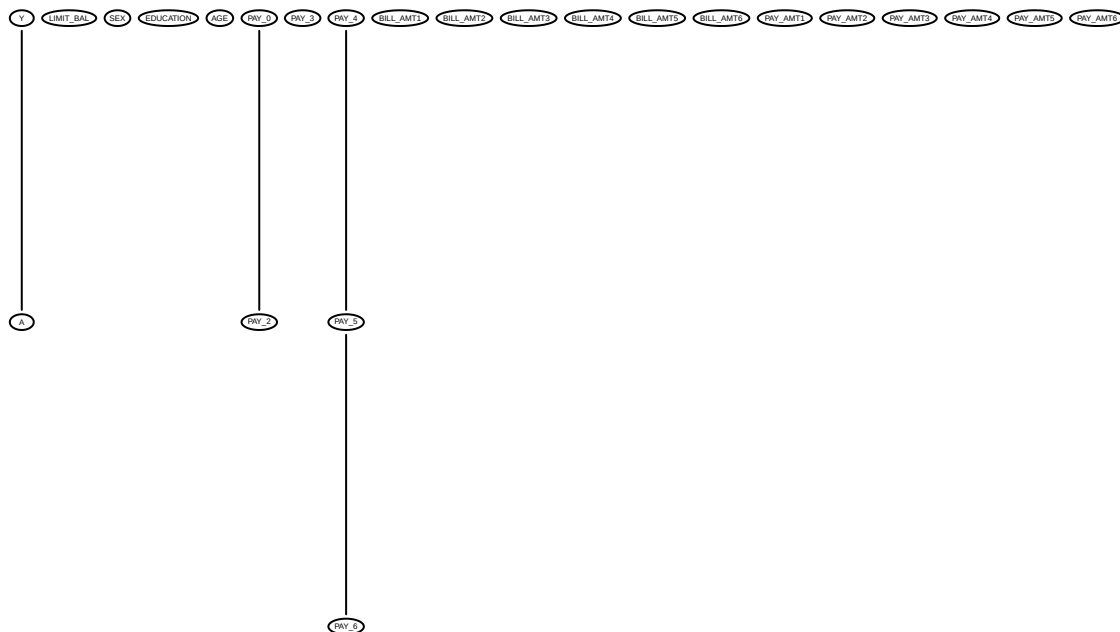
```
> # 3. Close the device to save the file
> dev.off()
```

pdf  
2

```
> # # Plot DAG
> # graphviz.plot(pc_fit_cextend, shape = "ellipse",
> # main = "Full Causal DAG, PC Method")
>
> # # 2. Export a high-quality version
> # png("OUTPUT_IMAGES_FROM_ANALYSIS_V6_10/causal_dag_pc_method_bnlearn_cextend.png", width = 1600, height = 1000)
> # graphviz.plot(pc_fit_cextend, shape = "ellipse",
> # main = "Full Causal DAG, PC Method")
> # 3. Close the device to save the file
> # dev.off()
```

```
> # Plot DAG
> graphviz.plot(pc_fit_bnlearn_maxsx, shape = "ellipse",
+ main = "Full Causal DAG, PC Method")
```

## Full Causal DAG, PC Method



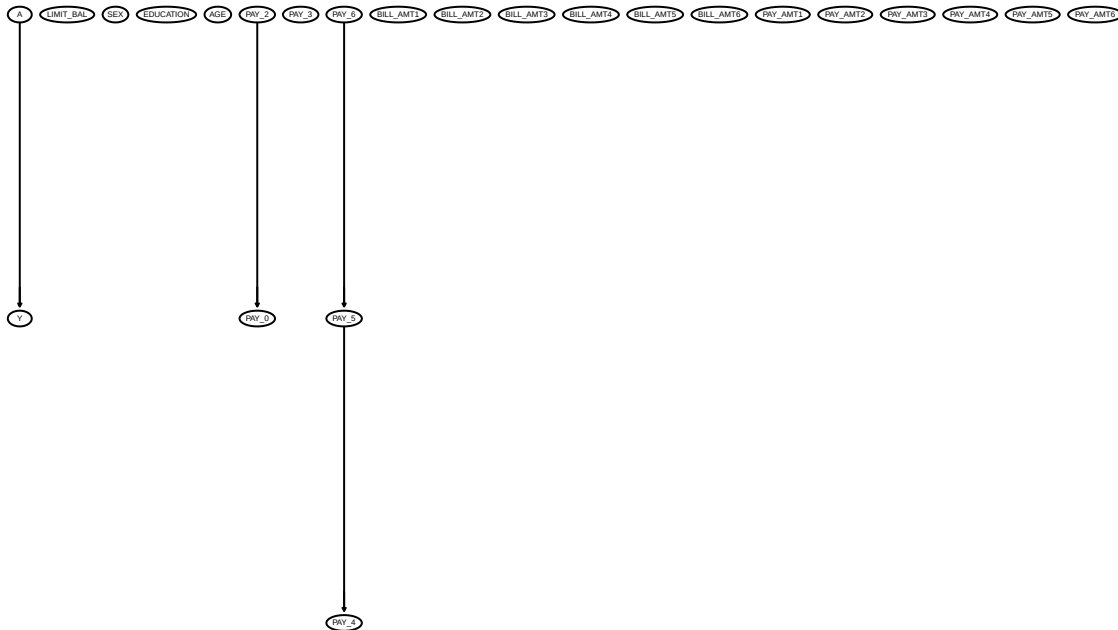
```
> # 2. Export a high-quality version
> png("OUTPUT_IMAGES_FROM_ANALYSIS_V6_10/causal_dag_pc_method_bnlearn_maxsx_test.png", width = 1600, height = 1000)
> graphviz.plot(pc_fit_bnlearn_maxsx, shape = "ellipse",
+ main = "TEST Full Causal DAG, PC Method")
> # 3. Close the device to save the file
> dev.off()
```

pdf  
2



```
> # Plot DAG
> graphviz.plot(pc_fit_bnlearn_maxsx_cextend, shape = "ellipse",
+ main = "TEST Full Causal DAG, PC Method")
```

## TEST Full Causal DAG, PC Method

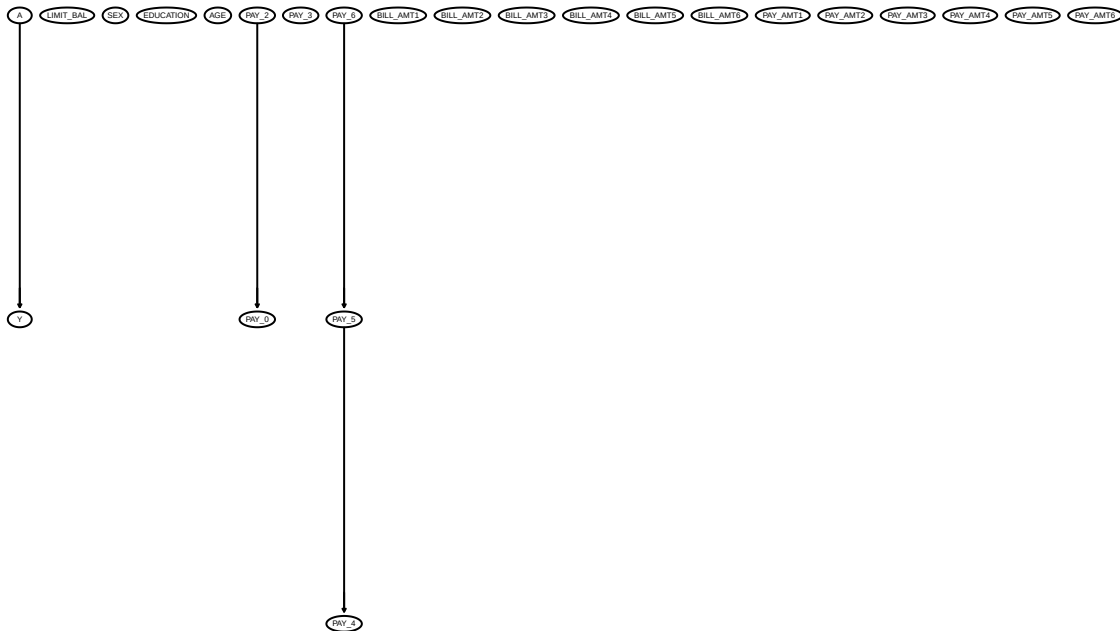


```
> # 2. Export a high-quality version
> png("OUTPUT_IMAGES_FROM_ANALYSIS_V6_10/causal_dag_pc_method_bnlearn_cextend_maxsx_test.png", width = 1000, height = 1000)
> graphviz.plot(pc_fit_bnlearn_maxsx_cextend, shape = "ellipse",
+ main = "TEST Full Causal DAG, PC Method")
> # 3. Close the device to save the file
> dev.off()
```

pdf  
2

```
> # Plot DAG
> graphviz.plot(pc_fit_bnlearn_maxsx_cextend, shape = "ellipse",
+ main = "TEST Full Causal DAG, PC Method")
```

## TEST Full Causal DAG, PC Method



```

> # 2. Export a high-quality version
> png("OUTPUT_IMAGES_FROM_ANALYSIS_V6_10/causal_dag_pc_method_bnlearn_cextend_maxsx_test.png", width = 1000, height = 1000)
> graphviz.plot(pc_fit_bnlearn_maxsx_cextend, shape = "ellipse",
+ main = "TEST Full Causal DAG, PC Method")
> # 3. Close the device to save the file
> dev.off()

```

pdf  
2

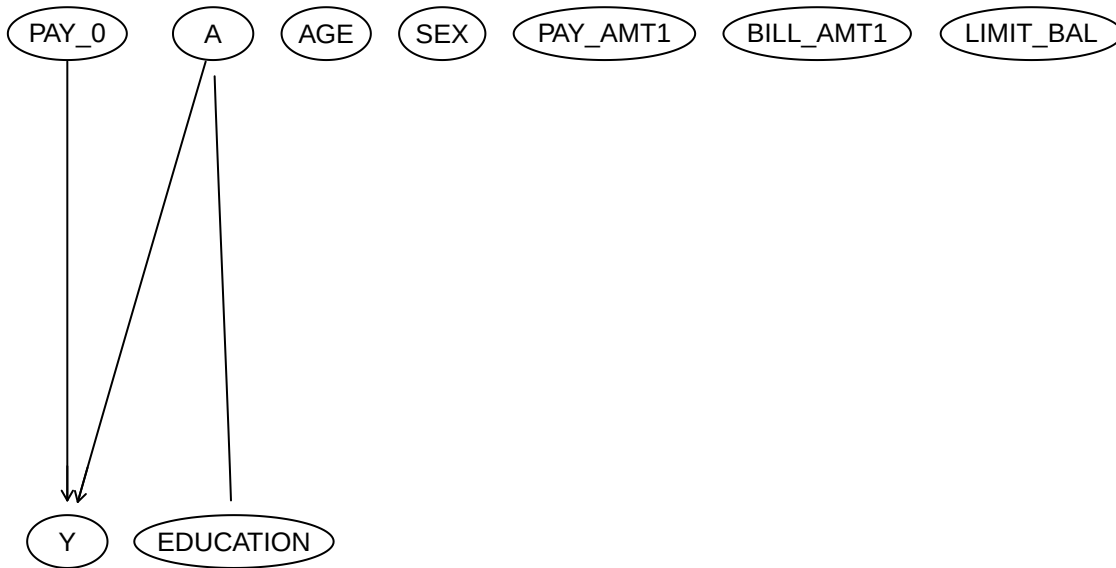
Since we see this isn't working well and the DAG is very sparse, let's try just limiting to the most recent variables, which should have stronger relationships with Y and thus be easier for the PC algorithm to find edges.

```

> # Plot DAG
> graphviz.plot(pc_fit_bnlearn_limitedcov, shape = "ellipse",
+ main = "Causal DAG: PC Method, Limited Covariates")

```

## Causal DAG: PC Method, Limited Covariates



```

> # 2. Export a high-quality version
> png("OUTPUT_IMAGES_FROM_ANALYSIS_V6_10/causal_dag_pc_method_bnlearn_limitedcov.png", width = 1600, height = 800)
> graphviz.plot(pc_fit_bnlearn_limitedcov, shape = "ellipse",
+ main = "Causal DAG: PC Method, Limited Covariates")
> # 3. Close the device to save the file
> dev.off()

```

```
pdf
2
```

```

> # Find indices of variables connected to Y
> pc_y_indices <- bnlearn::nbr(pc_fit_bnlearn_limitedcov, "Y") #all neighbors - but not spouses
> print(pc_y_indices)

```

```
[1] "A" "PAY_0"
```

```

> #parents, children, spouses using cextend version of the graph, which orients some edges and thus all relationships
> pc_y_indices_parents <- bnlearn::parents(pc_fit_bnlearn_limitedcov, "Y")
> pc_y_indices_children <- bnlearn::children(pc_fit_bnlearn_limitedcov, "Y")
> pc_y_indices_spouses <- bnlearn::spouses(pc_fit_bnlearn_limitedcov, "Y")
> print("All relationships identified by PC")

```

```
[1] "All relationships identified by PC"
```

```
> print(pc_y_indices_children)
```

```
character(0)
```

```
> print(pc_y_indices_parents)
```

```
[1] "A" "PAY_0"
```

```
> print(pc_y_indices_spouses)
```

```
character(0)
```

```

> pc_y_indices_all <- unique(c(pc_y_indices_parents, pc_y_indices_children, pc_y_indices_spouses, pc_y_
> print("All variables in Markov Blanket of Y identified by PC:")

[1] "All variables in Markov Blanket of Y identified by PC:"
> print(pc_y_indices_all)

[1] "A" "PAY_0"
> # Update your covariate list for the models
> cov_pc <- setdiff(pc_y_indices_all, "A") #exclude A
> print(cov_pc) #only PAY_0?

[1] "PAY_0"
> pc_a_indices <- bnlearn::nbr(pc_fit_bnlearn_limitedcov, "A") #all neighbors - but not spouses
> print(pc_a_indices)

[1] "Y" "EDUCATION"
> #parents, children, spouses using cextend version of the graph, which orients some edges and thus all
> pc_a_indices_parents <- bnlearn::parents(pc_fit_bnlearn_limitedcov, "A")
> pc_a_indices_children <- bnlearn::children(pc_fit_bnlearn_limitedcov, "A")
> pc_a_indices_spouses <- bnlearn::spouses(pc_fit_bnlearn_limitedcov, "A")
> print("All relationships identified by PC")

[1] "All relationships identified by PC"
> print(pc_a_indices_children)

[1] "Y"
> print(pc_a_indices_parents)

character(0)
> print(pc_a_indices_spouses)

[1] "PAY_0"
> pc_a_indices_all <- unique(c(pc_a_indices_parents, pc_a_indices_children, pc_a_indices_spouses, pc_a_
> print("All variables in Markov Blanket of A identified by PC:")

[1] "All variables in Markov Blanket of A identified by PC:"
> print(pc_a_indices_all)

[1] "Y" "PAY_0" "EDUCATION"
> cov_pc_a <-setdiff(pc_a_indices_all, "Y") #exclude Y, since we don't want to condition on the outcome
> print("Variables selected by PC (Markov Blanket of A):")

[1] "Variables selected by PC (Markov Blanket of A):"
> print(cov_pc_a)

[1] "PAY_0" "EDUCATION"

```

## 19 pc algorithm nontransformed or

```
> #outcome regression
> if(length(cov_pc) == 0) {
+ formula_or_pc <- as.formula("Y ~ A")
+ } else {
+ formula_or_pc <- as.formula(paste("Y ~ A +", paste(cov_pc, collapse = "+")))
+ }
> model_or_pc <- glm(formula_or_pc, data = data, family = binomial)
>
> print("formula")
```

```
[1] "formula"
```

```
> print(formula_or_pc)
```

```
Y ~ A + PAY_0
```

```
> # pc_data_all_enrolled <- data_all_enrolled %>%
> # # mutate(across(everything(), as.numeric)) %>%
> # select(Y, A, all_of(cov_all))
>
> # pc_data_no_enrolled <- data_no_enrolled %>%
> # # mutate(across(everything(), as.numeric)) %>%
> # select(Y, A, all_of(cov_all))
>
> # Predict default probability for world of everyone enrolled, and no one enrolled
> data$prob_Y_if_A1_pc_var <- predict(model_or_pc, newdata = data_all_enrolled, type = "response") #type
>
> EY_a1_or_pc_var <- mean(data$prob_Y_if_A1_pc_var)
>
> data$prob_Y_if_A0_pc_var <- predict(model_or_pc, newdata = data_no_enrolled, type = "response")
> EY_a0_or_pc_var <- mean(data$prob_Y_if_A0_pc_var)
>
> ace_or_pc_var <- EY_a1_or_pc_var - EY_a0_or_pc_var
>
> #print means
> print("Default likelihood (PC Set, OR), A=1 then A=0")
```

```
[1] "Default likelihood (PC Set, OR), A=1 then A=0"
```

```
> print(EY_a1_or_pc_var)
```

```
[1] 0.4284831
```

```
> print(EY_a0_or_pc_var)
```

```
[1] 0.2210747
```

```
> print(paste("OR ACE (PC Set):", round(ace_or_pc_var,4)))
```

```
[1] "OR ACE (PC Set): 0.2074"
```

## 20 PC algorithm nonttransformed ipw

```

> if(length(cov_pc_a) == 0) {
+ formula_ps_pc_var <- as.formula("A ~ 1")
+ } else {
+ formula_ps_pc_var <- as.formula(paste("A~", paste(cov_pc_a, collapse = "+"))) #formula of A~pc covar
+ }
> #get propensity scores
> model_ps_pc_var <- glm(formula_ps_pc_var, data = data, family = binomial) #binomial because only 2 ou
>
> print("formula")

[1] "formula"
> print(formula_ps_pc_var)

A ~ PAY_0 + EDUCATION
> data$ps_pc_var <- predict(model_ps_pc_var, type = "response") #calculate propensity scores for all cu
> print("min and max ps")

[1] "min and max ps"
> print(min(data$ps_pc_var))

[1] 4.721032e-07
> print(max(data$ps_pc_var))

[1] 0.9999995
> #extreme ps means likely have extreme scores
>
> data$ps_pc_var_clamped <- pmax(pmin(data$ps_pc_var, 0.9999), 0.0001)
>
>
> # calculate weights.
> data$weights_pc_var <- ifelse(data[["A"]] == 1, 1 / data$ps_pc_var_clamped, 1 / (1 - data$ps_pc_var_c
>
> print("min and max weights")

[1] "min and max weights"
> print(min(data$weights_pc_var)) #1.0001

[1] 1.0001
> print(max(data$weights_pc_var)) #2.8

[1] 7.066148
> #if customer was actually enrolled (A==1), then d weight as 1/ps, otherwise do 1/(1-ps)
> #So if customer had A=1 & ps=0.8, then weight = 1.25; but if A=0 then weight=5. Scales by unexpectedn
>
> # Estimate ACE using the weighted average
> #filter to only people who were enrolled A=1 or A=0
> a0_group <- data[data$A == 0,]
> a1_group <- data[data$A == 1,]
> # Filter to A0 or A1 groups, get default and multiply by weight; basically, if paid on time, contribu
> EY_a0_ipw_pc_var <- sum(as.numeric(as.character(a0_group$Y)) * a0_group$weights_pc_var)/sum(a0_group$
> EY_a1_ipw_pc_var <- sum(as.numeric(as.character(a1_group$Y)) * a1_group$weights_pc_var)/sum(a1_group$

```

```
>
> ace_ipw_pc_var <- EY_a1_ipw_pc_var - EY_a0_ipw_pc_var
> print(paste("IPW ACE (PC set):", round(ace_ipw_pc_var,4)))
```

```
[1] "IPW ACE (PC set): 0.2064"
```

## 21 Pc algorithm bootstrap nontransformed

```
> bootstrap_results_pc <- replicate(bootstrap_sample_size, {
+ #bootstrap sample
+ sample_df <- data[sample(nrow(data), replace = TRUE),]
+
+ #convert data to numerical for gaussian tests
+ sample_df_pc_data <- sample_df %>%
+ mutate(across(where(is.integer), as.numeric)) %>% #convert all integers to numeric, to run
+ select(Y, A, all_of(cov_all))
+
+ cov_most_recent <- c("PAY_0", "EDUCATION", "AGE", "SEX", "PAY_AMT1", "BILL_AMT1", "LIMIT_BAL")
+
+ sample_df_pc_data_limitedcov <- sample_df_pc_data %>%
+ select(Y, A, all_of(cov_most_recent))
+
+ pc_fit_boot <- pc.stable(sample_df_pc_data_limitedcov,
+ # blacklist = master_blacklist,
+ # test = "mix", #not needed, will automatically detect which test based on vari
+ alpha = 0.10)
+
+ adj_matrix <- bnlearn::amat(pc_fit_boot)
+
+ #extract markov blanket for y
+ y_indices_boot <- bnlearn::nbr(pc_fit_boot, "Y") #all neighbors - but not spouses
+
+ #parents, children, spouses using cextend version of the graph, which orients some edges and thus a
+ y_indices_parents_boot <- bnlearn::parents(pc_fit_boot, "Y")
+ y_indices_children_boot <- bnlearn::children(pc_fit_boot, "Y")
+ y_indices_spouses_boot <- bnlearn::spouses(pc_fit_boot, "Y")
+
+ y_indices_all_boot <- unique(c(y_indices_parents_boot, y_indices_children_boot, y_indices_spouses_b
+
+ # Update covariate list for the models
+ cov_pc_boot <- setdiff(y_indices_all_boot, "A")
+
+ #now for a
+ a_indices_boot <- bnlearn::nbr(pc_fit_boot, "A") #all neighbors - but not spouses
+ print(a_indices_boot)
+
+ #parents, children, spouses using cextend version of the graph, which orients some edges and thus a
+ a_indices_parents_boot <- bnlearn::parents(pc_fit_boot, "A")
+ a_indices_children_boot <- bnlearn::children(pc_fit_boot, "A")
+ a_indices_spouses_boot <- bnlearn::spouses(pc_fit_boot, "A")
+
+ a_indices_all_boot <- unique(c(a_indices_parents_boot, a_indices_children_boot, a_indices_spouses_b
```

```

+
+ cov_pc_boot_a <- setdiff(a_indices_all_boot, "Y")
+
+ # what if the blanket is empty? or
+ if(length(cov_pc_boot) == 0) {
+ formula_or_pc_boot <- as.formula("Y ~ A")
+ } else {
+ formula_or_pc_boot <- as.formula(paste("Y ~ A +", paste(cov_pc_boot, collapse = "+")))
+ }
+
+ #ipw
+ if(length(cov_pc_boot_a) == 0) {
+ formula_ps_pc_boot <- as.formula("A ~ 1")
+ } else {
+ formula_ps_pc_boot <- as.formula(paste("A ~", paste(cov_pc_boot_a, collapse = "+")))
+ }
+
+ # outcome regression
+ model_or_pc_boot <- glm(formula_or_pc_boot, data = sample_df, family = binomial)
+
+ #create fake datasets
+ sample_df_a0 <- sample_df
+ sample_df_a0$A <- factor(0, levels = levels(sample_df$A))
+ sample_df_a1 <- sample_df
+ sample_df_a1$A <- factor(1, levels = levels(sample_df$A))
+
+ # Predict default probability for world of everyone enrolled, and no one enrolled
+ sample_df_prob_Y_if_A1 <- predict(model_or_pc_boot, newdata = sample_df_a1, type = "response")
+ EY_a1_or_sample_df <- mean(sample_df_prob_Y_if_A1)
+
+ sample_df_prob_Y_if_A0 <- predict(model_or_pc_boot, newdata = sample_df_a0, type = "response")
+ EY_a0_or_sample_df <- mean(sample_df_prob_Y_if_A0)
+
+ ace_or_sample_df <- EY_a1_or_sample_df - EY_a0_or_sample_df
+
+ #IPW
+ #get propensity score
+ model_ps_sample_df <- glm(formula_ps_pc_boot, data = sample_df, family = binomial)
+ sample_df$ps_sample_df <- predict(model_ps_sample_df, type = "response")
+ #clamp ps score between 0.0001 and 0.9999 because otherwise get warnings of fitted prob being 0 or 1
+ sample_df$ps_sample_df_clamped <- pmax(pmin(sample_df$ps_sample_df, 0.9999), 0.0001)
+ #get weights
+ sample_df$weights_sample_df <- ifelse(sample_df$A == 1, 1/sample_df$ps_sample_df_clamped, 1/(1-sample_df$ps_sample_df_clamped))
+ #divide into a0 and a1 groups
+ a0_group_sample_df <- sample_df[sample_df$A == 0,]
+ a1_group_sample_df <- sample_df[sample_df$A == 1,]
+ #calculate ACE
+ EY_a0_ipw_sample_df <- sum(as.numeric(as.character(a0_group_sample_df$Y))) * a0_group_sample_df$weights_sample_df
+ EY_a1_ipw_sample_df <- sum(as.numeric(as.character(a1_group_sample_df$Y))) * a1_group_sample_df$weights_sample_df
+ ace_ipw_sample_df <- EY_a1_ipw_sample_df - EY_a0_ipw_sample_df
+
+ return(list(ace_or = ace_or_sample_df,
+ ace_ipw = ace_ipw_sample_df,
+ adj_matrix = adj_matrix,

```



```

+ blanket = y_indices_all_boot,
+ blanket_a = a_indices_all_boot
+))
+ }, simplify = FALSE)

```

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```

> orig_ace_pc_var <- c(ace_or_pc_var, ace_ipw_pc_var)
>
> #extract ACE values
> boot_aces_or_pc <- unlist(lapply(bootstrap_results_pc, function(x) x$ace_or))
> boot_aces_ipw_pc <- unlist(lapply(bootstrap_results_pc, function(x) x$ace_ipw))
>
> #extract markov blanket
> boot_blankets_pc <- lapply(bootstrap_results_pc, function(x) x$blanket)
> boot_blankets_pc_a <- lapply(bootstrap_results_pc, function(x) x$blanket_a)
>
> # convert to dataframe
> boot_df_pc_var <- data.frame(
+ Regression = boot_aces_or_pc,
+ IPW = boot_aces_ipw_pc
+)
> colnames(boot_df_pc_var) <- c("Regression", "IPW")
>
> # Calculate Metrics
> metrics_pc_var <- data.frame(
+ Method = c("Regression", "IPW"),
+ Original_ACE = orig_ace_pc_var, # Use values from the full dataset
+ Bootstrap_ACE = c(mean(boot_df_pc_var$Regression), mean(boot_df_pc_var$IPW)),
+ Standard_Error = c(sd(boot_df_pc_var$Regression), sd(boot_df_pc_var$IPW))
+)
>
> # Calculate Bias
> metrics_pc_var$Bootstrap_Bias <- metrics_pc_var$Bootstrap_ACE - metrics_pc_var$Original_ACE
> print(metrics_pc_var)

```

	Method	Original_ACE	Bootstrap_ACE	Standard_Error	Bootstrap_Bias
1	Regression	0.2074085	0.2086571	0.04027280	0.001248633
2	IPW	0.2063976	0.2008781	0.04347751	-0.005519497

```

> boot_blankets_pc_unlisted <- unlist(boot_blankets_pc)
>
> boot_mb_stability_pc <- table(boot_blankets_pc_unlisted) / bootstrap_sample_size
> print("Selection Frequency for Y's Markov Blanket:")

```

```
[1] "Selection Frequency for Y's Markov Blanket:"
```

```
> print(sort(boot_mb_stability_pc, decreasing = TRUE))
```

```
boot_blankets_pc_unlisted
 PAY_0 A EDUCATION SEX
 0.993 0.960 0.245 0.104
```

```

> boot_blankets_pc_a_unlisted <- unlist(boot_blankets_pc_a)
>
> boot_mb_stability_pc_a <- table(boot_blankets_pc_a_unlisted) / bootstrap_sample_size
> print("Selection Frequency for A's Markov Blanket:")

```

```
[1] "Selection Frequency for A's Markov Blanket:"
```

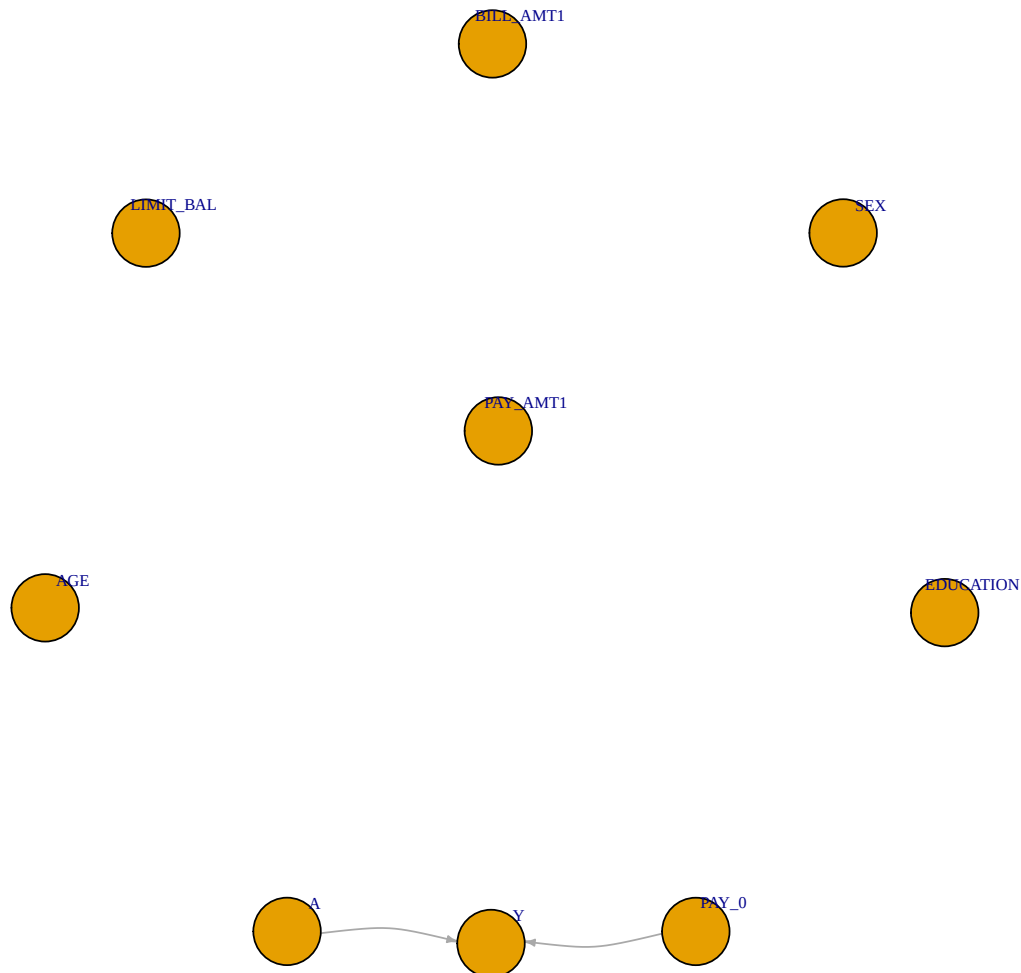
```
> print(sort(boot_mb_stability_pc_a, decreasing = TRUE))
```

```
boot_blankets_pc_a_unlisted
 Y PAY_0 EDUCATION SEX AGE LIMIT_BAL
```

0.960      0.676      0.473      0.110      0.001      0.001

```
> #get adjacent matrices
> all_adj_bootstrap_pc <- lapply(bootstrap_results_pc, function(x) x$adj_matrix)
> #sum them up to get edge frequencies - prob an edge exists between 2 variables
> consensus_matrix_bootstrap_pc <- Reduce("+", all_adj_bootstrap_pc) / bootstrap_sample_size #reduce sum
>
> # Create a graph where an edge exists if it appeared in > 50% of samples
> average_adj_bootstrap_pc <- (consensus_matrix_bootstrap_pc > 0.5) * 1
>
> # Convert back to a graph object for plotting
> avg_graph_bootstrap_pc <- graph_from_adjacency_matrix(average_adj_bootstrap_pc, mode = "directed")
>
> plot(avg_graph_bootstrap_pc,
+ main = "Consensus DAG (Edges in >50% of Bootstraps): PC Method, Limited Covariates",
+ vertex.size = 15, # Smaller nodes
+ vertex.label.cex = 0.6, # Smaller labels
+ vertex.label.dist = 1.2, # Offset labels from node centers
+ edge.arrow.size = 0.3, # Smaller arrows
+ edge.curved = 0.2) # Slight curve to distinguish overlapping arrows
```

## Consensus DAG (Edges in >50% of Bootstraps): PC Method, Limited Covariates



```

> png("OUTPUT_IMAGES_FROM_ANALYSIS_V6_10/consensus_dag_pc_bootstrap_limitedcov.png", width = 1200, height = 800)
> par(mar = c(1, 1, 3, 1))
> plot(avg_graph_bootstrap_pc,
+ main = "Consensus DAG (Edges in >50% of Bootstraps): PC Method, Limited Covariates",
+ vertex.size = 15, # Smaller nodes
+ vertex.label.cex = 0.6, # Smaller labels
+ vertex.label.dist = 1.2, # Offset labels from node centers
+ edge.arrow.size = 0.3, # Smaller arrows
+ edge.curved = 0.2) # Slight curve to distinguish overlapping arrows
> dev.off()

```

pdf  
2

## 22 6 Now the Analysis, for Local discovery methods

Use MMPC from bnlearn. The max-min parents and children algorithm. This searches the neighborhood of Y (parents and children) by testing for conditional independence at the  $\alpha=0.10$  level. Variable is only kept if relationship with Y can't be explained away by other variables.

## 23 nontransformed, local method

IAMB (Incremental Association)

The most common “Gold Standard” for medium-sized datasets.

How it works: It is dynamic. In every step of the growth phase, it re-tests all remaining variables and

The Pros: Much more reliable at finding the correct parents, children, and spouses than GS.

The Cons: Slower than GS (it runs  $O(n^2)$  tests).

N=500 Performance: Recommended. This is the most balanced choice for your credit data.

## 24 Test IAMB

```
> #convert data to numerical
> local_data_test <- data %>%
+ mutate(across(where(is.integer), as.numeric)) %>% #convert all integers to numeric, to run
+ select(Y, A, all_of(cov_all))
>
> local_data_test_mbonly <- local_data_test %>%
+ mutate(across(everything(), function(x) {
+ if(is.factor(x) || is.character(x)) as.numeric(as.factor(x)) else rank(x)
+ }))
>
> # Run nbr to find the markov blanket of Y only
> local_y_test <- learn.mb(local_data_test_mbonly, node = "Y", method = "iamb", alpha = 0.10, test="cor")
>
> print("Variables in the Local Neighborhood (Parents/Children) of Y:")
```

```
[1] "Variables in the Local Neighborhood (Parents/Children) of Y:"
```

```
> print(local_y_test)
```

```
[1] "PAY_0" "A" "PAY_3" "PAY_2" "PAY_AMT1"
```

```
> # Extract adjustment set (exclude treatment A)
> cov_local_test <- setdiff(local_y_test, "A") #L
> print(cov_local_test)
```

```
[1] "PAY_0" "PAY_3" "PAY_2" "PAY_AMT1"
```

```
> #now do the same for A to get covariates for propensity score model
> local_a_test <- learn.mb(local_data_test_mbonly, node = "A", method = "iamb", alpha = 0.10, test="cor")
> print("Variables in the Local Neighborhood (Parents/Children) of A:")
```

```
[1] "Variables in the Local Neighborhood (Parents/Children) of A:"
```

```
> print(local_a_test)
```

```
[1] "AGE" "Y" "EDUCATION" "LIMIT_BAL" "SEX"
```

```
> cov_local_a_test <- setdiff(local_a_test, "Y") #exclude Y, since we don't want to condition on the out
> print(cov_local_a_test)
```

```
[1] "AGE" "EDUCATION" "LIMIT_BAL" "SEX"
```

```

> # learn.mb(local_data, node = "A", method = "a", alpha = 0.10)
> #Error in check.label(algorithm, choices = ok, labels = learning.labels, : valid learning algorithm

> #convert data to numerical
> local_data <- data %>%
+ mutate(across(where(is.integer), as.numeric)) %>% #convert all integers to numeric, to run
+ select(Y, A, all_of(cov_all))
>
> local_data_test_mbonly <- local_data %>%
+ mutate(across(everything(), function(x) {
+ if(is.factor(x) || is.character(x)) as.numeric(as.factor(x)) else rank(x)
+ }))
>
>
> # Run nbr to find the markov blanket of Y only
> local_y <- learn.mb(local_data_test_mbonly, node = "Y", method = "inter.iamb", alpha = 0.10, test = "
>
> print("Variables in the Local Neighborhood (Parents/Children) of Y:")

[1] "Variables in the Local Neighborhood (Parents/Children) of Y:"
> print(local_y)

[1] "PAY_0" "A" "PAY_3" "PAY_2" "PAY_AMT1"
> # Extract adjustment set (exclude treatment A)
> cov_local <- setdiff(local_y, "A") #L
> print(cov_local)

[1] "PAY_0" "PAY_3" "PAY_2" "PAY_AMT1"
> #now do the same for A to get covariates for propensity score model
> local_a <- learn.mb(local_data_test_mbonly, node = "A", method = "inter.iamb", alpha = 0.10, test = "
> print("Variables in the Local Neighborhood (Parents/Children) of A:")

[1] "Variables in the Local Neighborhood (Parents/Children) of A:"
> print(local_a)

[1] "AGE" "Y" "EDUCATION" "LIMIT_BAL" "SEX"
> cov_local_a <- setdiff(local_a, "Y") #exclude Y, since we don't want to condition on the outcome in th
> print(cov_local_a)

[1] "AGE" "EDUCATION" "LIMIT_BAL" "SEX"

```

## 25 local algorithm nontransformed or

```

> # --- Outcome Regression (Local Set) ---
> if (length(cov_local) == 0) {
+ formula_or_local <- as.formula("Y ~ A")
+ } else {
+ formula_or_local <- as.formula(paste("Y ~ A +", paste(cov_local, collapse = " + ")))
+ }
>
> model_or_local <- glm(formula_or_local, data = data, family = binomial)
>

```

```

> # Counterfactual predictions
> EY_a1_or_local <- mean(predict(model_or_local, newdata = data_all_enrolled, type = "response"))
> EY_a0_or_local <- mean(predict(model_or_local, newdata = data_no_enrolled, type = "response"))
> ace_or_local <- EY_a1_or_local - EY_a0_or_local
>
> print(paste("Local Discovery OR ACE:", round(ace_or_local, 4)))

```

```
[1] "Local Discovery OR ACE: 0.205"
```

## 26 local algorithm nontransformed ipw

```

> if (length(cov_local_a) == 0) {
+ formula_ps_local <- as.formula("A ~ 1")
+ } else {
+ formula_ps_local <- as.formula(paste("A ~", paste(cov_local_a, collapse = " + ")))
+ }
>
> model_ps_local <- glm(formula_ps_local, data = data, family = binomial)
> data$ps_local <- predict(model_ps_local, type = "response")
> data$ps_local_clamped <- pmax(pmin(data$ps_local, 0.9999), 0.0001)
>
> # Weights
> data$weights_local <- ifelse(data$A == 1, 1 / data$ps_local_clamped, 1 / (1 - data$ps_local_clamped))
>
> a0_group <- data[data$A == 0,]
> a1_group <- data[data$A == 1,]
>
>
> EY_a0_ipw_local <- sum(as.numeric(as.character(a0_group$Y)) * a0_group$weights_local) / sum(a0_group$weights_local)
> EY_a1_ipw_local <- sum(as.numeric(as.character(a1_group$Y)) * a1_group$weights_local) / sum(a1_group$weights_local)
>
> ace_ipw_local <- EY_a1_ipw_local - EY_a0_ipw_local
> print(paste("Local Discovery IPW ACE:", round(ace_ipw_local, 4)))

```

```
[1] "Local Discovery IPW ACE: 0.2"
```

## 27 local algorithm nontransformed bootstrap

```

> bootstrap_results_local <- replicate(bootstrap_sample_size, {
+ sample_df <- data[sample(nrow(data), replace = TRUE),]
+
+ sample_df_local <- sample_df %>%
+ mutate(across(where(is.integer), as.numeric)) %>% #convert all integers to numeric, to run
+ select(Y, A, all_of(cov_all))
+
+ # This maps factors to integers and continuous variables to ranks
+ sample_df_ranked <- sample_df_local %>%
+ mutate(across(everything(), function(x) {
+ if(is.factor(x) || is.character(x)) as.numeric(as.factor(x)) else rank(x)
+ })))
+
+

```

```

+ # We use sample_df_ranked, method = "inter.iamb", and test = "cor"
+ local_y_boot <- tryCatch({
+ learn.mb(sample_df_ranked, node = "Y", method = "inter.iamb", test = "cor", alpha = 0.10)
+ }, error = function(e) character(0))
+
+ cov_local_boot <- setdiff(local_y_boot, "A")
+
+ local_a_boot <- tryCatch({
+ learn.mb(sample_df_ranked, node = "A", method = "inter.iamb", test = "cor", alpha = 0.10)
+ }, error = function(e) character(0))
+
+ cov_local_boot_a <- setdiff(local_a_boot, "Y")
+
+
+ if (length(cov_local_boot) == 0) {
+ formula_or_local_boot <- as.formula("Y ~ A")
+ } else {
+ formula_or_local_boot <- as.formula(paste("Y ~ A +", paste(cov_local_boot, collapse = " + ")))
+ }
+
+ if (length(cov_local_boot_a) == 0) {
+ formula_ps_local_boot <- as.formula("A ~ 1")
+ } else {
+ formula_ps_local_boot <- as.formula(paste("A ~", paste(cov_local_boot_a, collapse = " + ")))
+ }
+
+
+ # Uses sample_df so the GLM correctly processes the original factors/scales
+ model_or_local_boot <- glm(formula_or_local_boot, data = sample_df, family = binomial)
+
+ # create fake datasets
+ sample_df_a0 <- sample_df
+ sample_df_a0$A <- factor(0, levels = levels(sample_df$A))
+ sample_df_a1 <- sample_df
+ sample_df_a1$A <- factor(1, levels = levels(sample_df$A))
+
+ sample_df_prob_Y_if_A1 <- predict(model_or_local_boot, newdata = sample_df_a1, type = "response")
+ EY_a1_or_sample_df <- mean(sample_df_prob_Y_if_A1)
+
+ sample_df_prob_Y_if_A0 <- predict(model_or_local_boot, newdata = sample_df_a0, type = "response")
+ EY_a0_or_sample_df <- mean(sample_df_prob_Y_if_A0)
+
+ ace_or_sample_df <- EY_a1_or_sample_df - EY_a0_or_sample_df
+
+ # IPW
+ model_ps_local_boot <- glm(formula_ps_local_boot, data = sample_df, family = binomial)
+ sample_df$ps_sample_df <- predict(model_ps_local_boot, type = "response")
+ sample_df$ps_sample_df_clamped <- pmax(pmin(sample_df$ps_sample_df, 0.9999), 0.0001)
+ sample_df$weights_sample_df <- ifelse(sample_df$A == 1, 1/sample_df$ps_sample_df_clamped, 1/(1-sample_df$ps_sample_df_clamped))
+
+ a0_group_sample_df <- sample_df[sample_df$A == 0,]
+ a1_group_sample_df <- sample_df[sample_df$A == 1,]
+
+ EY_a0_ipw_sample_df <- sum(as.numeric(as.character(a0_group_sample_df$Y))) * a0_group_sample_df$weights_sample_df

```

```

+ EY_a1_ipw_sample_df <- sum(as.numeric(as.character(a1_group_sample_df$Y)) * a1_group_sample_df$weight)
+
+ ace_ipw_sample_df <- EY_a1_ipw_sample_df - EY_a0_ipw_sample_df
+
+ return(list(ace_or = ace_or_sample_df,
+ ace_ipw = ace_ipw_sample_df,
+ blanket = local_y_boot,
+ blanket_a = local_a_boot
+))
+ }, simplify=FALSE)

```

```

> orig_ace_local_var <- c(ace_or_local, ace_ipw_local)
>
> #extract ACE values
> boot_aces_or_local <- unlist(lapply(bootstrap_results_local, function(x) x$ace_or))
> boot_aces_ipw_local <- unlist(lapply(bootstrap_results_local, function(x) x$ace_ipw))
>
> #extract markov blanket
> boot_blankets_local <- lapply(bootstrap_results_local, function(x) x$blanket)
> boot_blankets_local_a <- lapply(bootstrap_results_local, function(x) x$blanket_a)
>
>
> # convert to dataframe
> boot_df_local_var <- data.frame(
+ Regression = boot_aces_or_local,
+ IPW = boot_aces_ipw_local
+)
> colnames(boot_df_local_var) <- c("Regression", "IPW")
>
> # Calculate Metrics
> metrics_local_var <- data.frame(
+ Method = c("Regression", "IPW"),
+ Original_ACE = orig_ace_local_var,
+ Bootstrap_ACE = c(mean(boot_df_local_var$Regression), mean(boot_df_local_var$IPW)),
+ Standard_Error = c(sd(boot_df_local_var$Regression), sd(boot_df_local_var$IPW))
+)
>
> # Calculate Bias
> metrics_local_var$Bootstrap_Bias <- metrics_local_var$Bootstrap_ACE - metrics_local_var$Original_ACE
> print(metrics_local_var)

```

	Method	Original_ACE	Bootstrap_ACE	Standard_Error	Bootstrap_Bias
1	Regression	0.2049753	0.2022124	0.04198828	-0.002762913
2	IPW	0.2000219	0.2063077	0.04732621	0.006285757

```

> boot_blankets_local_unlisted <- unlist(boot_blankets_local)
>
> boot_mb_stability_local <- table(boot_blankets_local_unlisted) / bootstrap_sample_size
> print("Selection Frequency for Y's Markov Blanket:")

```

```
[1] "Selection Frequency for Y's Markov Blanket:"
```

```
> print(sort(boot_mb_stability_local, decreasing = TRUE))
```

```
boot_blankets_local_unlisted
```



A	PAY_0	PAY_3	PAY_2	EDUCATION	BILL_AMT1	PAY_AMT2	PAY_AMT3
1.000	0.992	0.713	0.494	0.373	0.362	0.287	0.285
PAY_AMT4	AGE	PAY_AMT1	BILL_AMT6	LIMIT_BAL	PAY_AMT5	PAY_6	BILL_AMT2
0.285	0.265	0.232	0.225	0.174	0.113	0.100	0.095
SEX	PAY_5	PAY_4	BILL_AMT3	BILL_AMT5	PAY_AMT6	BILL_AMT4	
0.094	0.084	0.080	0.076	0.075	0.075	0.046	

```
> boot_blankets_local_unlisted_a <- unlist(boot_blankets_local_a)
>
> boot_mb_stability_local_a <- table(boot_blankets_local_unlisted_a) / bootstrap_sample_size
> print("Selection Frequency for A's Markov Blanket:")
```

```
[1] "Selection Frequency for A's Markov Blanket:"
```

```
> print(sort(boot_mb_stability_local_a, decreasing = TRUE))
```

boot_blankets_local_unlisted_a							
AGE	Y	EDUCATION	SEX	PAY_0	LIMIT_BAL	PAY_AMT4	PAY_AMT2
1.000	0.998	0.915	0.535	0.370	0.326	0.245	0.242
BILL_AMT2	BILL_AMT1	PAY_AMT6	PAY_AMT3	BILL_AMT6	BILL_AMT5	PAY_2	PAY_AMT1
0.233	0.222	0.188	0.167	0.137	0.127	0.116	0.110
PAY_3	PAY_4	PAY_AMT5	PAY_6	BILL_AMT3	PAY_5	BILL_AMT4	
0.104	0.102	0.090	0.089	0.076	0.073	0.052	

## 28 7 Now for the iamb with blacklist

```
> allowed_nodes_local_forbidden_edges <- c("Y", "A", cov_all)
> # Nothing can cause Demographics
> bl_demog <- expand.grid(from = allowed_nodes_local_forbidden_edges, to = c("AGE", "SEX", "EDUCATION"))
>
> # Y cannot cause anything
> bl_y <- expand.grid(from = "Y", to = setdiff(allowed_nodes_local_forbidden_edges, "Y"))
>
> # A cannot cause anything except Y
> bl_a <- expand.grid(from = "A", to = setdiff(allowed_nodes_local_forbidden_edges, c("A", "Y")))
>
> # temporal relationships: later months cannot cause earlier months
> pay_vars <- c("PAY_0", "PAY_2", "PAY_3", "PAY_4", "PAY_5", "PAY_6")
> bill_amt_vars <- c("BILL_AMT1", "BILL_AMT2", "BILL_AMT3", "BILL_AMT4", "BILL_AMT5", "BILL_AMT6")
> pay_amt_vars <- c("PAY_AMT1", "PAY_AMT2", "PAY_AMT3", "PAY_AMT4", "PAY_AMT5", "PAY_AMT6")
>
> bl_temporal <- data.frame()
> for(i in 1:(length(pay_vars)-1)) {
+ for(j in (i+1):length(pay_vars)) {
+ # Block Present -> Past
+ bl_temporal <- rbind(bl_temporal, data.frame(from = pay_vars[i], to = pay_vars[j]))
+ bl_temporal <- rbind(bl_temporal, data.frame(from = bill_amt_vars[i], to = bill_amt_vars[j]))
+ bl_temporal <- rbind(bl_temporal, data.frame(from = pay_amt_vars[i], to = pay_amt_vars[j]))
+ }
+ }
>
> # Combine all into one master blacklist
> master_blacklist <- rbind(bl_demog, bl_y, bl_a, bl_temporal) %>%
+ mutate(across(everything(), as.character)) %>% # Convert everything to character to avoid factor le
```

```

+ filter(from != to) %>% # Remove self-loops
+ distinct() # Remove duplicates
>
> print("Master Blacklist:")

```

```
[1] "Master Blacklist:"
```

```
> print(master_blacklist)
```

	from	to
1	Y	AGE
2	A	AGE
3	LIMIT_BAL	AGE
4	SEX	AGE
5	EDUCATION	AGE
6	PAY_0	AGE
7	PAY_2	AGE
8	PAY_3	AGE
9	PAY_4	AGE
10	PAY_5	AGE
11	PAY_6	AGE
12	BILL_AMT1	AGE
13	BILL_AMT2	AGE
14	BILL_AMT3	AGE
15	BILL_AMT4	AGE
16	BILL_AMT5	AGE
17	BILL_AMT6	AGE
18	PAY_AMT1	AGE
19	PAY_AMT2	AGE
20	PAY_AMT3	AGE
21	PAY_AMT4	AGE
22	PAY_AMT5	AGE
23	PAY_AMT6	AGE
24	Y	SEX
25	A	SEX
26	LIMIT_BAL	SEX
27	EDUCATION	SEX
28	AGE	SEX
29	PAY_0	SEX
30	PAY_2	SEX
31	PAY_3	SEX
32	PAY_4	SEX
33	PAY_5	SEX
34	PAY_6	SEX
35	BILL_AMT1	SEX
36	BILL_AMT2	SEX
37	BILL_AMT3	SEX
38	BILL_AMT4	SEX
39	BILL_AMT5	SEX
40	BILL_AMT6	SEX
41	PAY_AMT1	SEX
42	PAY_AMT2	SEX
43	PAY_AMT3	SEX
44	PAY_AMT4	SEX

45	PAY_AMT5	SEX
46	PAY_AMT6	SEX
47	Y	EDUCATION
48	A	EDUCATION
49	LIMIT_BAL	EDUCATION
50	SEX	EDUCATION
51	AGE	EDUCATION
52	PAY_0	EDUCATION
53	PAY_2	EDUCATION
54	PAY_3	EDUCATION
55	PAY_4	EDUCATION
56	PAY_5	EDUCATION
57	PAY_6	EDUCATION
58	BILL_AMT1	EDUCATION
59	BILL_AMT2	EDUCATION
60	BILL_AMT3	EDUCATION
61	BILL_AMT4	EDUCATION
62	BILL_AMT5	EDUCATION
63	BILL_AMT6	EDUCATION
64	PAY_AMT1	EDUCATION
65	PAY_AMT2	EDUCATION
66	PAY_AMT3	EDUCATION
67	PAY_AMT4	EDUCATION
68	PAY_AMT5	EDUCATION
69	PAY_AMT6	EDUCATION
70	Y	A
71	Y	LIMIT_BAL
72	Y	PAY_0
73	Y	PAY_2
74	Y	PAY_3
75	Y	PAY_4
76	Y	PAY_5
77	Y	PAY_6
78	Y	BILL_AMT1
79	Y	BILL_AMT2
80	Y	BILL_AMT3
81	Y	BILL_AMT4
82	Y	BILL_AMT5
83	Y	BILL_AMT6
84	Y	PAY_AMT1
85	Y	PAY_AMT2
86	Y	PAY_AMT3
87	Y	PAY_AMT4
88	Y	PAY_AMT5
89	Y	PAY_AMT6
90	A	LIMIT_BAL
91	A	PAY_0
92	A	PAY_2
93	A	PAY_3
94	A	PAY_4
95	A	PAY_5
96	A	PAY_6
97	A	BILL_AMT1
98	A	BILL_AMT2

99	A	BILL_AMT3	
100	A	BILL_AMT4	
101	A	BILL_AMT5	
102	A	BILL_AMT6	
103	A	PAY_AMT1	
104	A	PAY_AMT2	
105	A	PAY_AMT3	
106	A	PAY_AMT4	
107	A	PAY_AMT5	
108	A	PAY_AMT6	
109	PAY_0		PAY_2
110	BILL_AMT1	BILL_AMT2	
111	PAY_AMT1	PAY_AMT2	
112	PAY_0		PAY_3
113	BILL_AMT1	BILL_AMT3	
114	PAY_AMT1	PAY_AMT3	
115	PAY_0		PAY_4
116	BILL_AMT1	BILL_AMT4	
117	PAY_AMT1	PAY_AMT4	
118	PAY_0		PAY_5
119	BILL_AMT1	BILL_AMT5	
120	PAY_AMT1	PAY_AMT5	
121	PAY_0		PAY_6
122	BILL_AMT1	BILL_AMT6	
123	PAY_AMT1	PAY_AMT6	
124	PAY_2		PAY_3
125	BILL_AMT2	BILL_AMT3	
126	PAY_AMT2	PAY_AMT3	
127	PAY_2		PAY_4
128	BILL_AMT2	BILL_AMT4	
129	PAY_AMT2	PAY_AMT4	
130	PAY_2		PAY_5
131	BILL_AMT2	BILL_AMT5	
132	PAY_AMT2	PAY_AMT5	
133	PAY_2		PAY_6
134	BILL_AMT2	BILL_AMT6	
135	PAY_AMT2	PAY_AMT6	
136	PAY_3		PAY_4
137	BILL_AMT3	BILL_AMT4	
138	PAY_AMT3	PAY_AMT4	
139	PAY_3		PAY_5
140	BILL_AMT3	BILL_AMT5	
141	PAY_AMT3	PAY_AMT5	
142	PAY_3		PAY_6
143	BILL_AMT3	BILL_AMT6	
144	PAY_AMT3	PAY_AMT6	
145	PAY_4		PAY_5
146	BILL_AMT4	BILL_AMT5	
147	PAY_AMT4	PAY_AMT5	
148	PAY_4		PAY_6
149	BILL_AMT4	BILL_AMT6	
150	PAY_AMT4	PAY_AMT6	
151	PAY_5		PAY_6
152	BILL_AMT5	BILL_AMT6	

## 29 nontransformed, local method

IAMB (Incremental Association)

The most common “Gold Standard” for medium-sized datasets.

How it works: It is dynamic. In every step of the growth phase, it re-tests all remaining variables and

The Pros: Much more reliable at finding the correct parents, children, and spouses than GS.

The Cons: Slower than GS (it runs  $O(n^2)$  tests).

N=500 Performance: Recommended. This is the most balanced choice for your credit data.

```
> #convert data to numerical
> local_data <- data %>%
+ mutate(across(where(is.integer), as.numeric)) %>% #convert all integers to numeric, to run
+ select(Y, A, all_of(cov_all))
>
> #because local discovery, need to exclude Y from blacklist...
> master_blacklist_excl_y <- master_blacklist %>%
+ filter(from != "Y" & to != "Y") %>%
+ as.matrix()
>
> master_blacklist_matrix_excl_y <- as.matrix(master_blacklist_excl_y)
>
> # Run nbr to find the markov blanket of Y only
> local_y_blacklist <- learn.mb(local_data, node = "Y", method = "inter.iamb", blacklist = master_blacklist)
>
> print("Variables in the Local Neighborhood (Parents/Children) of Y:")
```

```
[1] "Variables in the Local Neighborhood (Parents/Children) of Y:"
```

```
> print(local_y_blacklist)
```

```
character(0)
```

```
> # Extract adjustment set (exclude treatment A)
> cov_local_blacklist <- setdiff(local_y_blacklist, "A") #L
> print(cov_local_blacklist)
```

```
character(0)
```

```
> master_blacklist_excl_a <- master_blacklist %>%
+ filter(from != "A" & to != "A") %>%
+ as.matrix()
>
> master_blacklist_matrix_excl_a <- as.matrix(master_blacklist_excl_a)
>
>
> #now do the same for A to get covariates for propensity score model
> local_a_blacklist <- learn.mb(local_data, node = "A", method = "inter.iamb", blacklist = master_blacklist)
> print("Variables in the Local Neighborhood (Parents/Children) of A:")
```

```
[1] "Variables in the Local Neighborhood (Parents/Children) of A:"
```

```
> print(local_a_blacklist)
```

```
character(0)
```

```
> cov_local_a_blacklist <- setdiff(local_a_blacklist, "Y") #exclude Y, since we don't want to condition on Y
> print(cov_local_a_blacklist)
```

```
character(0)
```

### 30 local algorithm nontransformed or

```
> # --- Outcome Regression (Local Set) ---
> if (length(cov_local_blacklist) == 0) {
+ formula_or_local_blacklist <- as.formula("Y ~ A")
+ } else {
+ formula_or_local_blacklist <- as.formula(paste("Y ~ A +", paste(cov_local_blacklist, collapse = " + "
+ }
>
> model_or_local_blacklist <- glm(formula_or_local_blacklist, data = data, family = binomial)
>
> # Counterfactual predictions
> EY_a1_or_local_blacklist <- mean(predict(model_or_local_blacklist, newdata = data_all_enrolled, type = "response"))
> EY_a0_or_local_blacklist <- mean(predict(model_or_local_blacklist, newdata = data_no_enrolled, type = "response"))
> ace_or_local_blacklist <- EY_a1_or_local_blacklist - EY_a0_or_local_blacklist
>
> print(paste("Local Discovery OR ACE:", round(ace_or_local_blacklist, 4)))
```

```
[1] "Local Discovery OR ACE: 0.1869"
```

### 31 local algorithm nontransformed ipw

```
> if (length(cov_local_a_blacklist) == 0) {
+ formula_ps_local_blacklist <- as.formula("A ~ 1")
+ } else {
+ formula_ps_local_blacklist <- as.formula(paste("A ~", paste(cov_local_a_blacklist, collapse = " + "
+ }
>
> model_ps_local_blacklist <- glm(formula_ps_local_blacklist, data = data, family = binomial)
> data$ps_local_blacklist <- predict(model_ps_local_blacklist, type = "response")
> data$ps_local_clamped_blacklist <- pmax(pmin(data$ps_local_blacklist, 0.9999), 0.0001)
>
> # Weights
> data$weights_local_blacklist <- ifelse(data$A == 1, 1 / data$ps_local_clamped_blacklist, 1 / (1 - data$ps_local_clamped_blacklist))
>
> a0_group <- data[data$A == 0,]
> a1_group <- data[data$A == 1,]
>
>
> EY_a0_ipw_local_blacklist <- sum(as.numeric(as.character(a0_group$Y)) * a0_group$weights_local_blacklist)
> EY_a1_ipw_local_blacklist <- sum(as.numeric(as.character(a1_group$Y)) * a1_group$weights_local_blacklist)
>
> ace_ipw_local_blacklist <- EY_a1_ipw_local_blacklist - EY_a0_ipw_local_blacklist
```

```
> print(paste("Local Discovery IPW ACE:", round(ace_ipw_local_blacklist, 4)))

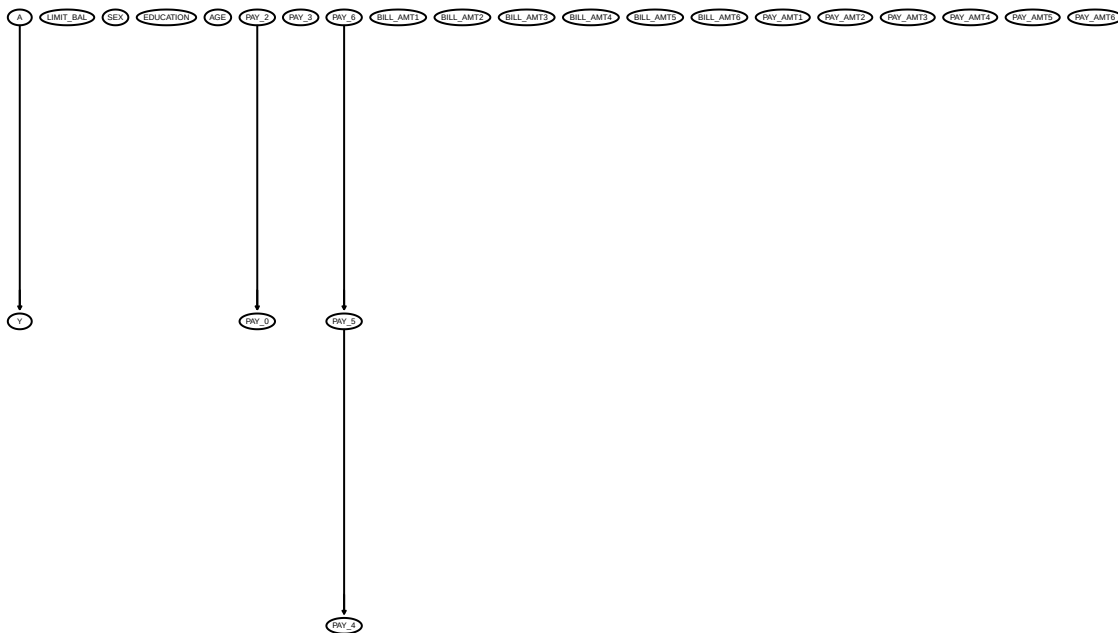
[1] "Local Discovery IPW ACE: 0.1869"

> orig_ace_local_var_blacklist <- c(ace_or_local_blacklist, ace_ipw_local_blacklist)
```

## 32 8 pc with only most recent month bnlearn, with blacklist

```
> #convert data to numerical for gaussian tests
> pc_data <- data %>%
+ mutate(across(where(is.integer), as.numeric)) %>% #convert all integers to numeric, to run
+ select(Y, A, all_of(cov_all))
>
> # calculate correlation matrix and sample size; use to test for conditional independence amongst pairs
> # pc_corr_size <- list(C = cor(pc_data), n = nrow(pc_data))
>
>
> pc_fit_bnlearn_blacklist <- pc.stable(pc_data,
+ blacklist = master_blacklist,
+ # test = "mix", #not needed, will automatically detect which test based on vari
+ alpha = 0.10)
>
>
> # Get adjacency matrix from PC fit
> # pc_fit_adj_matrix <- amat(pc_fit)
> # print(pc_fit_adj_matrix)
>
> pc_fit_bnlearn_cextend_blacklist <- cextend(pc_fit_bnlearn_blacklist) #directs edges as much as possi
>
>
> pc_fit_bnlearn_maxsx_blacklist <- pc.stable(pc_data,
+ blacklist = master_blacklist,
+ # test = "mix", #not needed, will automatically detect which test based on vari
+ alpha = 0.10,
+ max.sx=2) #limits size of conditioning set to 2, which should help with the power
> pc_fit_bnlearn_maxsx_cextend_blacklist <- cextend(pc_fit_bnlearn_maxsx_blacklist) #directs edges as m
>
> # Plot DAG
> graphviz.plot(pc_fit_bnlearn_blacklist, shape = "ellipse",
+ main = "TEST Full Causal DAG, PC Method")
```

## TEST Full Causal DAG, PC Method



```

> # 2. Export a high-quality version
> png("OUTPUT_IMAGES_FROM_ANALYSIS_V6_10/causal_dag_pc_method_bnlearn_blacklist_test.png", width = 1600
> graphviz.plot(pc_fit_bnlearn_blacklist, shape = "ellipse",
+ main = "TEST Full Causal DAG, PC Method")
> # 3. Close the device to save the file
> dev.off()

```

pdf  
2

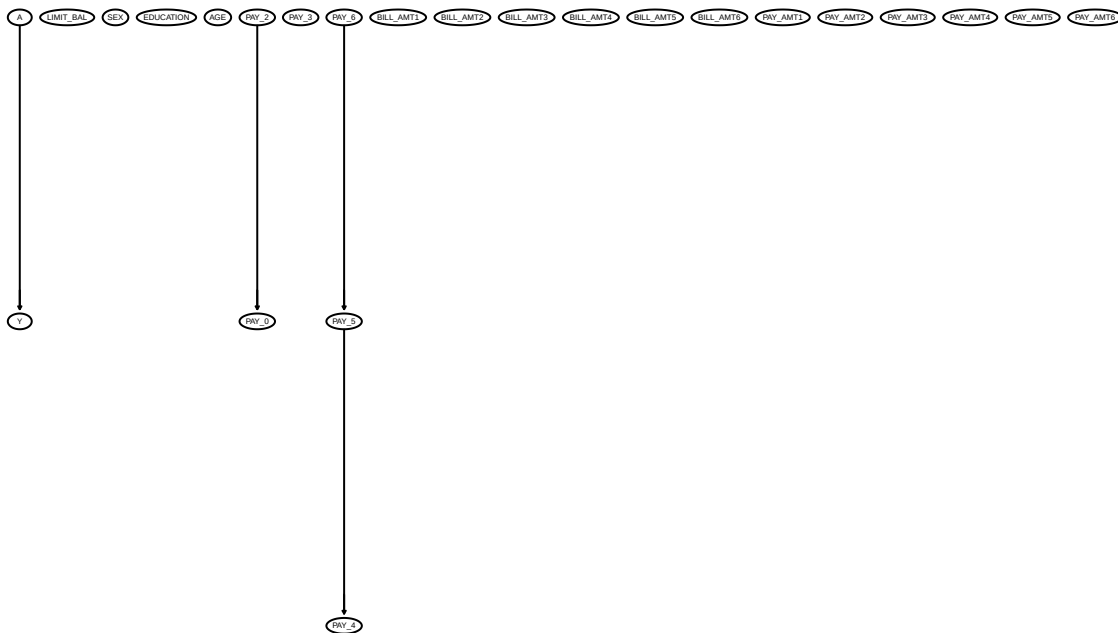
```

> # Plot DAG
> graphviz.plot(pc_fit_bnlearn_cextend_blacklist, shape = "ellipse",
+ main = "TEST Full Causal DAG, PC Method")

```



## TEST Full Causal DAG, PC Method



```

> # 2. Export a high-quality version
> png("OUTPUT_IMAGES_FROM_ANALYSIS_V6_10/causal_dag_pc_method_bnlearn_cextend_blacklist_test.png", width=1000, height=1000)
> graphviz.plot(pc_fit_bnlearn_cextend_blacklist, shape = "ellipse",
+ main = "TEST Full Causal DAG, PC Method")
> # 3. Close the device to save the file
> dev.off()

```

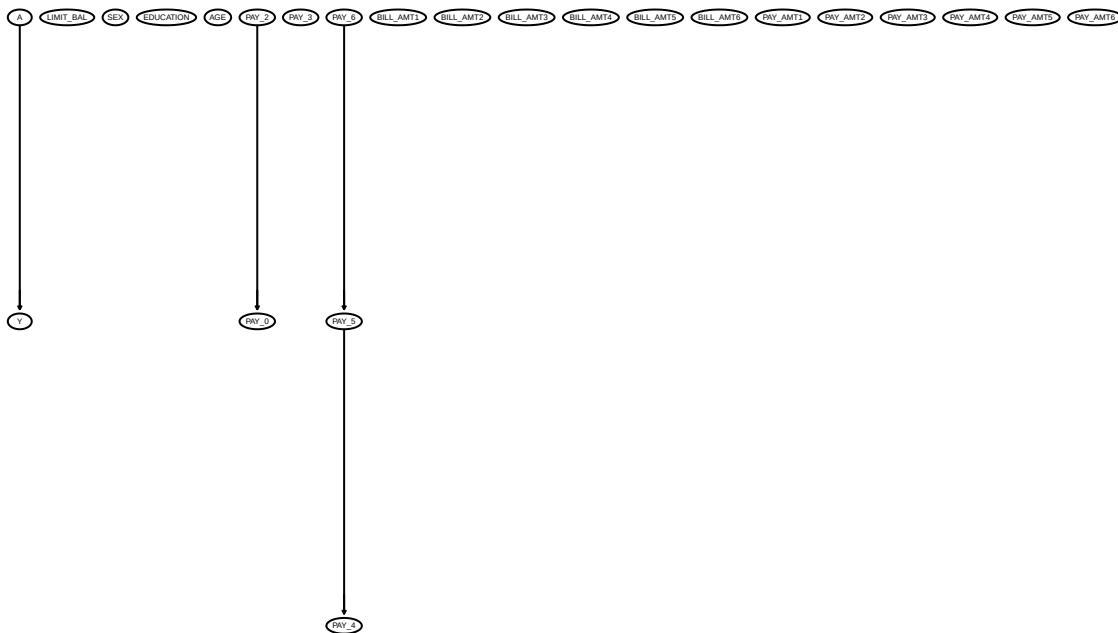
pdf  
2

```

> # Plot DAG
> graphviz.plot(pc_fit_bnlearn_maxsx_blacklist, shape = "ellipse",
+ main = "TESTFull Causal DAG, PC Method")

```

## TESTFull Causal DAG, PC Method



```

> # 2. Export a high-quality version
> png("OUTPUT_IMAGES_FROM_ANALYSIS_V6_10/causal_dag_pc_method_bnlearn_maxsx_blacklist_test.png", width = 1000, height = 1000)
> graphviz.plot(pc_fit_bnlearn_maxsx_blacklist, shape = "ellipse",
+ main = "TEST Full Causal DAG, PC Method")
> # 3. Close the device to save the file
> dev.off()

```

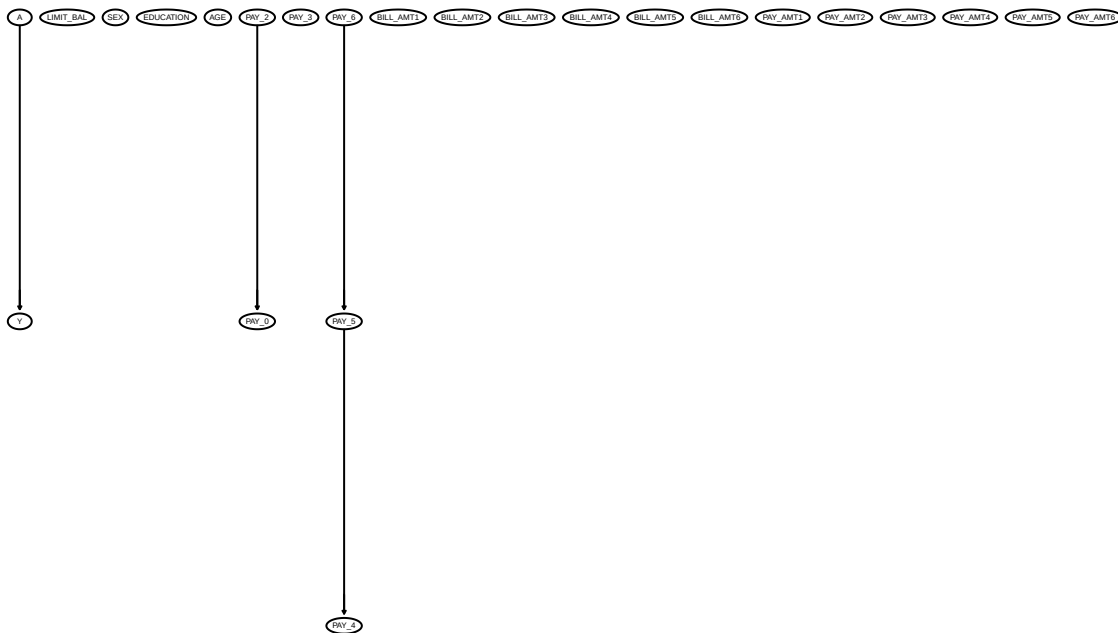
pdf  
2

```

> # Plot DAG
> graphviz.plot(pc_fit_bnlearn_maxsx_cextend_blacklist, shape = "ellipse",
+ main = "TEST Full Causal DAG, PC Method")

```

## TEST Full Causal DAG, PC Method



```
> # 2. Export a high-quality version
> png("OUTPUT_IMAGES_FROM_ANALYSIS_V6_10/causal_dag_pc_method_bnlearn_cextend_maxsx_blacklist_test.png")
> graphviz.plot(pc_fit_bnlearn_maxsx_cextend_blacklist, shape = "ellipse",
+ main = "TEST Full Causal DAG, PC Method")
> # 3. Close the device to save the file
> dev.off()
```

pdf  
2

This doesn't work very well, no point in blacklisting...

**33 9 end Note: compare the ACEs for all of the runs at the end!!  
maybe do a big table to discuss.**

```
> # Define a helper function to label and combine
> combine_metrics <- function(df, selection_name, data_type) {
+ df %>%
+ mutate(
+ Selection = selection_name,
+ Rel_Bias_Pct = ((Bootstrap_ACE - Original_ACE) / Original_ACE) * 100
+ # Bias_SE_Ratio = abs(Bootstrap_ACE - Original_ACE) / Standard_Error
+) %>%
+ mutate(across(where(is.numeric), ~round(.x, 4))) %>%
+ mutate(Rel_Bias_Pct = round(Rel_Bias_Pct, 2)) %>%
+ select(Selection, Method, Original_ACE, Bootstrap_ACE, everything())
+ }
>
> # Create a master table
```

```

> all_metrics_master <- bind_rows(
+ combine_metrics(metrics_all_var, "All Variables"),
+ combine_metrics(metrics_lasso, "Lasso"),
+ combine_metrics(metrics_pc_var, "PC, Limited Covariates"),
+ combine_metrics(metrics_rank_pc_var, "Rank PC"),
+ combine_metrics(metrics_local_var, "IAMB"),
+ # combine_metrics(metrics_pc_var_full, "PC, All Covariates")
+ # combine_metrics(metrics_local_var_blacklist, "IAMB with Blacklist")
+)
>
> print(all_metrics_master)

```

	Selection	Method	Original_ACE	Bootstrap_ACE	Standard_Error
1	All Variables	Regression	0.2064	0.2147	0.0439
2	All Variables	IPW	0.2256	0.2363	0.0632
3	Lasso	Regression	0.2033	0.2045	0.0401
4	Lasso	IPW	0.1783	0.1953	0.0428
5	PC, Limited Covariates	Regression	0.2074	0.2087	0.0403
6	PC, Limited Covariates	IPW	0.2064	0.2009	0.0435
7	Rank PC	Regression	0.2064	0.2029	0.0414
8	Rank PC	IPW	0.1827	0.1881	0.0467
9	IAMB	Regression	0.2050	0.2022	0.0420
10	IAMB	IPW	0.2000	0.2063	0.0473
Bootstrap_Bias Rel_Bias_Pct					
1			0.0083	4.03	
2			0.0107	4.73	
3			0.0012	0.58	
4			0.0170	9.55	
5			0.0012	0.60	
6			-0.0055	-2.67	
7			-0.0035	-1.69	
8			0.0054	2.96	
9			-0.0028	-1.35	
10			0.0063	3.14	

```

> # 1. Prepare the data (Rounding and Renaming)
> table_data <- all_metrics_master %>%
+ mutate(
+ # Match the decimal logic from your gt code
+ Original_ACE = round(Original_ACE, 4),
+ Bootstrap_ACE = round(Bootstrap_ACE, 4),
+ Standard_Error = round(Standard_Error, 4),
+ Bootstrap_Bias = round(Bootstrap_Bias, 4),
+ Rel_Bias_Pct = paste0(round(Rel_Bias_Pct, 1), "%") # Format as string for ggplot
+ # Bias_SE_Ratio = round(Bias_SE_Ratio, 3)
+) %>%
+ # Rename columns to match your desired header
+ rename(
+ `Estimation Method` = Method,
+ `Original ACE` = Original_ACE,
+ `Bootstrap Mean ACE` = Bootstrap_ACE,
+ `Bootstrap Standard Error (SE)` = Standard_Error,
+ `Bootstrap Bias` = Bootstrap_Bias,
+ `Relative Bootstrap Bias (%)` = Rel_Bias_Pct,

```

```

+ # `Bootstrap Bias-to-SE Ratio` = Bias_SE_Ratio
+)
>
> # 2. Create the table graphic
> # We use 'ttheme_minimal' or 'ttheme_clean' for a thesis look
> tab_plot <- ggtexttable(table_data,
+ rows = NULL,
+ theme = ttheme("classic", base_size = 10)) %>%
+ tab_add_title(text = "ACE Estimation Comparison Across Variable Selection Methods",
+ face = "bold", padding = unit(1.5, "line"))
>
> # 3. Save as PNG using ggsave (No Chrome needed!)
> ggsave(filename = "OUTPUT_IMAGES_FROM_ANALYSIS_V6_10/ACE_Comparison_Table.png",
+ plot = tab_plot,
+ width = 12, height = 8, dpi = 300, bg = "white")
>
> # View it in RStudio
> print(tab_plot)

```

## ison Across Variable Selection Methods

Estimation Method	Original ACE	Bootstrap Mean ACE	Bootstrap Standard Error (SE)	Bootstrap Bias
Regression	0.2064	0.2147	0.0439	0.0083
IPW	0.2256	0.2363	0.0632	0.0107
Regression	0.2033	0.2045	0.0401	0.0012
IPW	0.1783	0.1953	0.0428	0.0170
Regression	0.2074	0.2087	0.0403	0.0012
IPW	0.2064	0.2009	0.0435	-0.0055
Regression	0.2064	0.2029	0.0414	-0.0035
IPW	0.1827	0.1881	0.0467	0.0054
Regression	0.2050	0.2022	0.0420	-0.0028
IPW	0.2000	0.2063	0.0473	0.0063

```

> ## 1. Create the professional gt table object
> # final_table_img <- all_metrics_master %>%
> # gt(groupname_col = "Selection") %>% # Group by Selection method
> #
> # # Add title and subtitle for your thesis
> # tab_header(
> # title = md("ACE Estimation Comparison Across Variable Selection Methods"),
> #) %>%
> #
> # # Format column labels clearly
> # cols_label(
> # Method = md("Estimation Method"),
> # Original_ACE = md("Original ACE"),
> # Bootstrap_ACE = md("Bootstrap Mean ACE"),
> # Standard_Error = md("Bootstrap Standard Error (SE)"),

```

```

> # Rel_Bias_Pct = md("Relative Bootstrap Bias (%)"),
> # Bias_SE_Ratio = md("Bootstrap Bias-to-SE Ratio"),
> # Bootstrap_Bias = md("Bootstrap Bias")
> #) %>%
>
> # # Add formatting for scientific precision and percentages
> # fmt_number(columns = c(Original_ACE, Bootstrap_ACE, Standard_Error), decimals = 4) %>%
> # fmt_percent(columns = Rel_Bias_Pct, decimals = 1, scale_values=FALSE) %>%
> # fmt_number(columns = Bias_SE_Ratio, decimals = 3) %>%
>
> # # Choose a clean theme (like minimal or pff)
> # gt_theme_pff()
>
> # # 2. Save the table as an image (PNG)
> # # Ensure webshot2 is installed for this step
> # gtsave(final_table_img, filename = "OUTPUT_IMAGES_FROM_ANALYSIS_V6_10/ACE_Comparison_Table.png")
>
> # # Print the table to the Viewer to see it immediately
> # final_table_img

> print("Formula for ordinary regression model for all variables:")

[1] "Formula for ordinary regression model for all variables:"
> print(formula_or_all_var)

Y ~ A + LIMIT_BAL + SEX + EDUCATION + AGE + PAY_0 + PAY_2 + PAY_3 +
 PAY_4 + PAY_5 + PAY_6 + BILL_AMT1 + BILL_AMT2 + BILL_AMT3 +
 BILL_AMT4 + BILL_AMT5 + BILL_AMT6 + PAY_AMT1 + PAY_AMT2 +
 PAY_AMT3 + PAY_AMT4 + PAY_AMT5 + PAY_AMT6

> print("Formula for IPW model for all variables:")

[1] "Formula for IPW model for all variables:"
> print(formula_ps_all_var)

A ~ LIMIT_BAL + SEX + EDUCATION + AGE + PAY_0 + PAY_2 + PAY_3 +
 PAY_4 + PAY_5 + PAY_6 + BILL_AMT1 + BILL_AMT2 + BILL_AMT3 +
 BILL_AMT4 + BILL_AMT5 + BILL_AMT6 + PAY_AMT1 + PAY_AMT2 +
 PAY_AMT3 + PAY_AMT4 + PAY_AMT5 + PAY_AMT6

> print("Formula for ordinary regression model for lasso:")

[1] "Formula for ordinary regression model for lasso:"
> print("It gets cut off so I just output the covariates instead of a full formula as well")

[1] "It gets cut off so I just output the covariates instead of a full formula as well"
> print(formula_lasso_or_text)

[1] "Y ~ A1 + LIMIT_BAL + EDUCATION2 + EDUCATION4 + AGE + PAY_0-1 + PAY_00 + PAY_01 + PAY_02 + PAY_03 +
 PAY_04 + PAY_05 + PAY_06 + BILL_AMT1 + BILL_AMT2 + BILL_AMT3 + BILL_AMT4 + BILL_AMT5 + BILL_AMT6 +
 PAY_AMT1 + PAY_AMT2 + PAY_AMT3 + PAY_AMT4 + PAY_AMT5 + PAY_AMT6"

> print(formula_lasso_or_text_anotherver)

[1] "A1" "LIMIT_BAL" "EDUCATION2" "EDUCATION4" "AGE"
[6] "PAY_0-1" "PAY_00" "PAY_01" "PAY_02" "PAY_03"
[11] "PAY_06" "PAY_24" "PAY_25" "PAY_32" "PAY_34"

```

```

[16] "PAY_4-1" "PAY_42" "PAY_43" "PAY_54" "PAY_62"
[21] "BILL_AMT5" "PAY_AMT2" "PAY_AMT3" "PAY_AMT4" "PAY_AMT6"
> print("Formula for IPW model for lasso variable selection:")

[1] "Formula for IPW model for lasso variable selection:"
> print("It gets cut off so I just output the covariates instead of a full formula as well")

[1] "It gets cut off so I just output the covariates instead of a full formula as well"
> print(formula_lasso_ps_text)

[1] "A ~ SEX1 + EDUCATION3 + EDUCATION5 + AGE + PAY_04 + PAY_AMT6"
> print(formula_lasso_ps_text_anotherver)

[1] "SEX1" "EDUCATION3" "EDUCATION5" "AGE" "PAY_04"
[6] "PAY_AMT6"
> print("Formula for ordinary regression model for PC, limiting to most recent month:")

[1] "Formula for ordinary regression model for PC, limiting to most recent month:"
> print(formula_or_pc)

Y ~ A + PAY_0
> print("Formula for IPW for PC, limiting to most recent month:")

[1] "Formula for IPW for PC, limiting to most recent month:"
> print(formula_ps_pc_var)

A ~ PAY_0 + EDUCATION
> print("Formula for ordinary regression model for Rank PC:")

[1] "Formula for ordinary regression model for Rank PC:"
> print(formula_or_rank_pc)

Y ~ A + PAY_3 + PAY_0
> print("Formula for IPW for Rank PC:")

[1] "Formula for IPW for Rank PC:"
> print(formula_ps_rank_pc_var)

A ~ AGE + EDUCATION + PAY_3
> print("Formula for ordinary regression model for local discovery (INTER_IAMB)")

[1] "Formula for ordinary regression model for local discovery (INTER_IAMB)"
> print(formula_or_local)

Y ~ A + PAY_0 + PAY_3 + PAY_2 + PAY_AMT1
> print("formula for IPW for local discovery (INTER_IAMB)")

[1] "formula for IPW for local discovery (INTER_IAMB)"

```

```
> print(formula_ps_local)
```

```
A ~ AGE + EDUCATION + LIMIT_BAL + SEX
```

```
> # print("Formula for ordinary regression model for PC with ALL covariates")
```

```
> # print(formula_or_pc_full)
```

```
> # print("formula for IPW for PC with ALL covariates")
```

```
> # print(formula_ps_pc_var_full)
```

```
> # Calculate 95% Confidence Intervals using the Normal Approximation: ACE +/- 1.96 * SE
```

```
> all_metrics_master <- all_metrics_master %>%
```

```
+ mutate(
```

```
+ CI_lower = Bootstrap_ACE - (1.96 * Standard_Error),
```

```
+ CI_upper = Bootstrap_ACE + (1.96 * Standard_Error)
```

```
+)
```

```
> ace_se_graph <- ggplot(all_metrics_master, aes(x = Selection, y = Standard_Error, fill = Method)) +
```

```
+ geom_bar(stat = "identity", position = "dodge") +
```

```
+ scale_fill_manual(values = c("Regression" = "#2c7fb8", "IPW" = "#feb24c")) +
```

```
+ coord_flip() +
```

```
+ theme_minimal() +
```

```
+ labs(title = "Comparison of Model Variance (Standard Error)",
```

```
+ y = "Standard Error of ACE", x = "Variable Selection Method")+
```

```
+ theme(
```

```
+ legend.position = "bottom",
```

```
+ legend.text = element_text(size = 8), # Slightly smaller text to save space
```

```
+ legend.title = element_text(size = 9, face = "bold"),
```

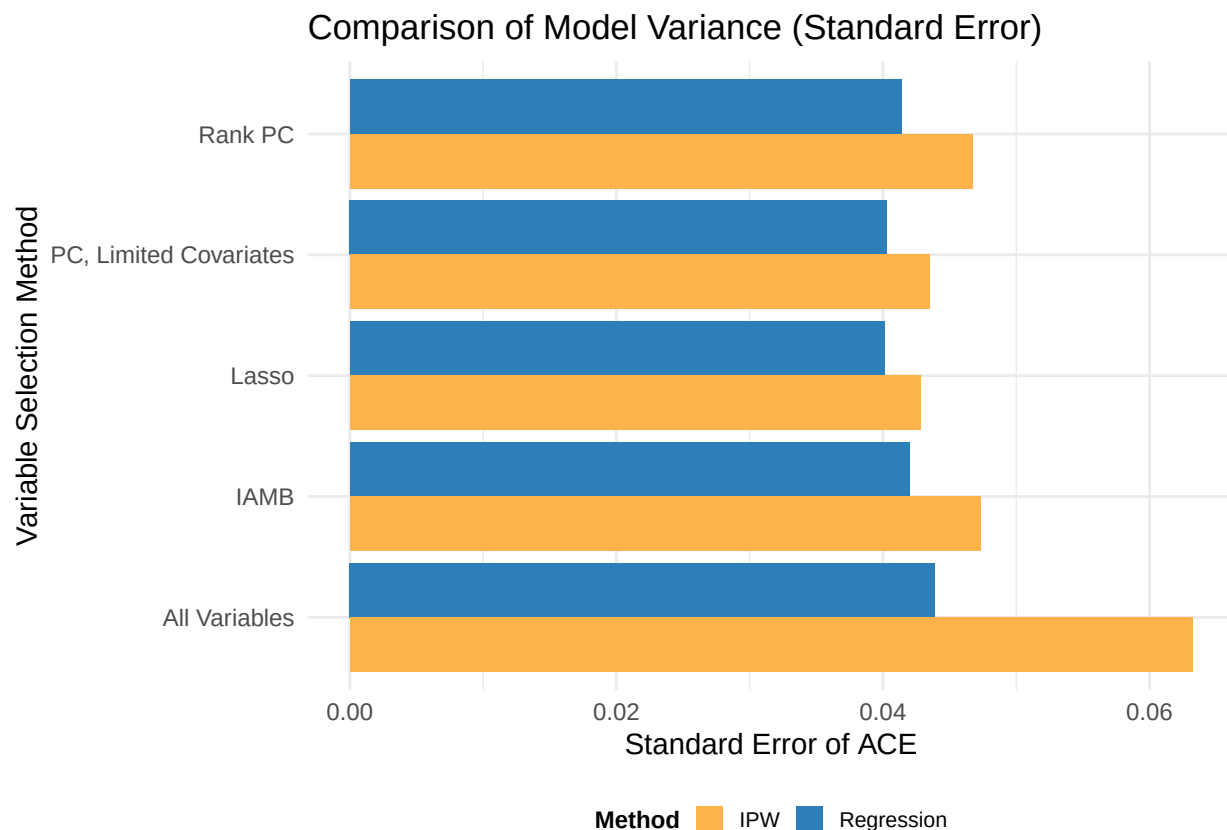
```
+ legend.key.size = unit(0.4, "cm") # Smaller keys to fit more on screen
```

```
+)
```

```
> ggsave("OUTPUT_IMAGES_FROM_ANALYSIS_V6_10/ace_standard_errors.png", plot = ace_se_graph, width = 8, height = 6)
```

```
> print(ace_se_graph)
```





```

> ace_ci_graph <- ggplot(all_metrics_master, aes(x = Selection, y = Bootstrap_ACE, color = Method, group = Selection)) +
+ geom_errorbar(aes(ymin = CI_lower, ymax = CI_upper),
+ position = position_dodge(width = 0.9),
+ width = 0.5, alpha = 0.6) +
+ geom_point(aes(y = Bootstrap_ACE),
+ position = position_dodge(width = 0.9),
+ size = 3) +
+ geom_point(aes(y = Original_ACE),
+ shape = 4,
+ position = position_dodge(width = 0.9),
+ size = 4, stroke = 1.2) +
+ coord_flip() +
+ theme_minimal() +
+ scale_color_manual(values = c("Regression" = "#2c7fb8", "IPW" = "#feb24c")) +
+ labs(title = "Average Causal Effect (ACE) Comparisons",
+ subtitle = "Dots (•) = Bootstrap Mean ACE (w/ 95% CI) | Crosses (x) = Original Estimate of ACE",
+ y = "Estimated ACE (Effect on Default Probability)",
+ x = "Variable Selection Strategy",
+ color = "Method",
+ shape = "Data Format") +
+ theme(
+ legend.position = "bottom",
+ panel.grid.major.y = element_line(color = "grey90") # Makes the rows easier to follow
+)
>
> ggsave("OUTPUT_IMAGES_FROM_ANALYSIS_V6_10/ace_confidence_intervals.png", plot = ace_ci_graph, width = 10, height = 10)
> print(ace_ci_graph)

```

## Average Causal Effect (ACE) Comparisons

Dots (.) = Bootstrap Mean ACE (w/ 95% CI) | Crosses (x) = Original Estir

